THE UNIVERSITY *of* EDINBURGH
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Senior Honours Project

Simulation-based inference in search for CP violating effects in diboson production at the TeV scale

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Abstract

In order to explain the observed asymmetry between matter and antimatter in the universe, additional sources of Charge Parity violation beyond the Standard Model are needed. To observe instances of CP violation, CP-odd observables must be found as CP-violation manifests in the interference between SM and CP-odd operators. This work used neural network models to construct a machine-learning based CP-odd observable in diboson production. The observable successfully demonstrated asymmetry in interference term distributions. The results of this work present additional proxies that can be used in the search for CP violation in diboson production.

Declaration

I declare that this project and report is my own work.

Signature: C. Looney Date: 28/11/2025

Supervisor: Dr. Dominik Duda 10 Weeks

Contents

1 Introduction

The Standard Model of particle physics (SM) has proved to be an extremely successful theory for describing the fundamental particles that are responsible for matter and forces [1, 2]. The discoveries of the W and Z bosons at the CERN proton-antiproton collider [3], and the discovery of the Higgs boson [4, 5] at the Large Hadron Collider (LHC) further validated the success of the SM. However, there are still several questions the SM is currently unable to answer: what causes the observed baryonic asymmetry [6]? What is dark matter [7]? What is the path towards the unification of forces [8]?

The observed baryonic asymmetry in the universe indicates the presence of charge-parity (CP) violation, one of the three Sakharov conditions required for baryogensis [9]. If matter and antimatter were governed by perfectly symmetrical laws, they would have been created in equal amounts in the early universe and subsequently annihilated, leaving behind a cosmos devoid of matter. While the SM contains some explanations of CP violation, such as the Cabibbo-Kobayashi-Maskawa mechanism [10, 11], the predicted amount is not enough to account for the observed asymmetry [12]. This inspires the search for CP violation of a greater scale in Beyond Standard Model (BSM) physics.

Extensions to the SM predict the existence of heavy resonances – peaks around certain energies in the differential cross-section of scattering experiments – that can decay into pairs of vector bosons (WW, WZ, and ZZ); this has been a recent experimental focus in the search for BSM physics [13]. However, the lack of detected resonances calls for a more general approach that indirectly searches for very heavy new physics rather than directly probing for particle existence [13, 14].

Effective Field Theories (EFT) are a powerful tool for generalizing contributions from BSM physics by considering the contribution of SM operators and BSM operators to the effective Lagrangian [15, 16]. This method allows virtual contributions from BSM physics to manifest as modifications to interaction amplitudes [15, 16]. While resonance searches target the production of new heavy states [17], CP-violating effects arise from higher-dimensional EFT operators that alter the amplitudes of interactions [15, 18, 19]. This work investigates the CP-odd interference contributions to diboson production at the TeV scale, corresponding to the proton-proton collision energy scale used in the search for diboson resonances [13, 17, 20].

Certain dimension-six Standard Model Effective Field Theory (SMEFT) operators introduce CP-odd modifications to diboson production [18, 19, 21, 22, 23], producing interference contributions in production amplitudes. This work employs simulation-based inference to investigate this interference, leveraging machine learning to determine CP-sensitive observables. Machine learning provides the ability to combine many weakly CP violation sensitive observables into a single discriminant.

Two distinct neural network architectures are developed and compared: a Deep Neural Network (DNN) trained on high-level, reconstructed kinematic variables, and a Graph Neural Network (GNN) that operates directly on low-level constituent information to exploit the geometric and kinematic correlations within the event. As CP-odd operators only produce small contributions to the interference term, the observable signatures of CP violation are expected to be extremely subtle. The goal of this work is not to achieve strong classification performance, but to extract measurable asymmetries in a machine-learning produced observable.

2 Background

2.1 The Standard Model

The Standard Model (SM) describes the fundamental particles and their interactions. Exchange particles within the SM are responsible for the forces that govern our universe; for example, the photon mediates the electromagnetic force and the gluon mediates the strong force. As demonstrated in Figure 1, the particle content of the SM can be divided into three sections: Quarks, Leptons (matter), and Bosons (mediators of the forces).

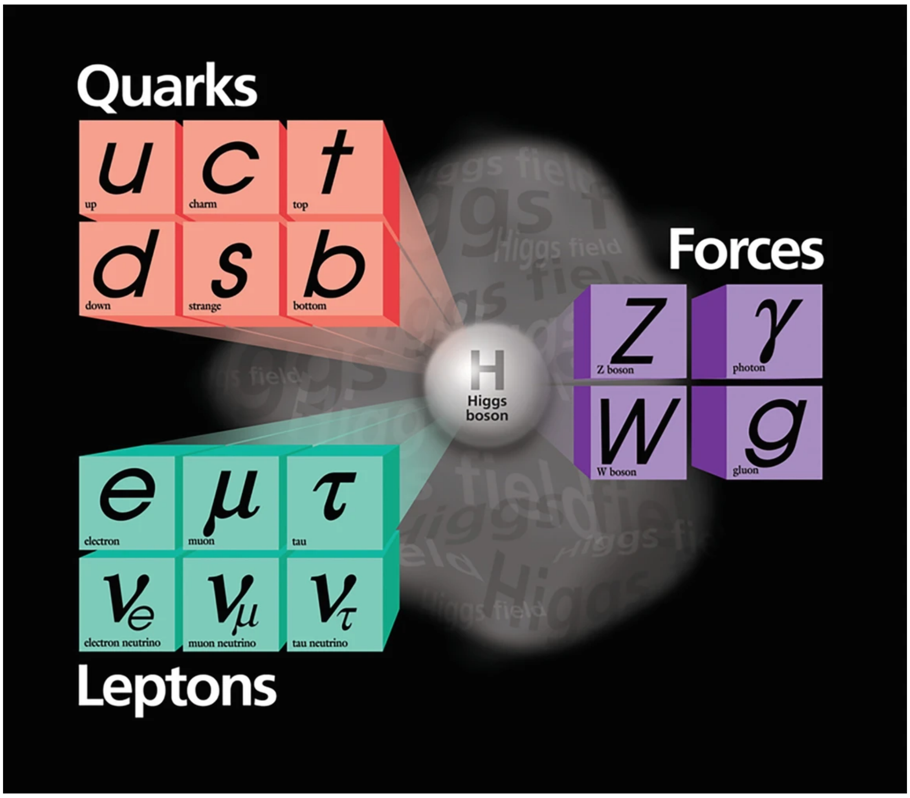


Figure 1: Particle content within the Standard Model [24]

The W and Z bosons, which mediate the weak interaction, were first observed in the early 1980s at CERN, providing experimental evidence in favour of the unified description of the electromagnetic and weak forces [3]. In this work, the signal process considered is electroweak diboson production. The dominant contribution is quark-antiquark annihilation,

$$q\bar{q} \to W^* \to WZ \tag{1}$$

Where the intermediate W^* is produced off-shell and subsequently a Z boson. A schematic

Feynman diagram representing this processes is shown in Figure 2.

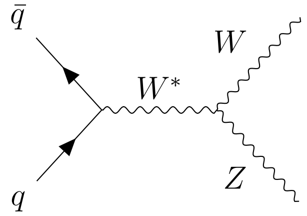


Figure 2: Leading-order diboson production process: $q\bar{q} \to W^* \to WZ$.

For the production of diboson pairs, three semileptonic final-states, in which one boson decays leptonically and the other hadronically, are available: $ZV \to \nu\nu qq$, $WV \to \ell\nu qq$, and $ZV \to \ell\ell qq$ [13]. This work investigates the semileptonic final-state: $\ell\ell qq$. This final-state is chosen due to its high signal-to-background ratio compared to fully hadronic and fully leptonic final-states [17].

Several extensions to the SM predict the existence of heavy diboson resonances, leading to several experiments searching for these resonances in high-energy proton-proton scattering events [13, 17]. These heavy diboson resonances can be created through several processes: gluon-gluon fusion, Drell-Yan, or vector-boson fusion as demonstrated in Figure 3 [13]. The absence of detections indicates that the production of heavy diboson resonances requires energy levels exceeding that of current colliders [13, 14]. Therefore, a more general approach is required which indirectly measures virtual contributions from BSM physics rather than directly probing for the new particles [13, 14]. This points the current direction of research to EFT, which allows for the virtual contributions to manifest as alterations to interaction amplitudes [15, 16].

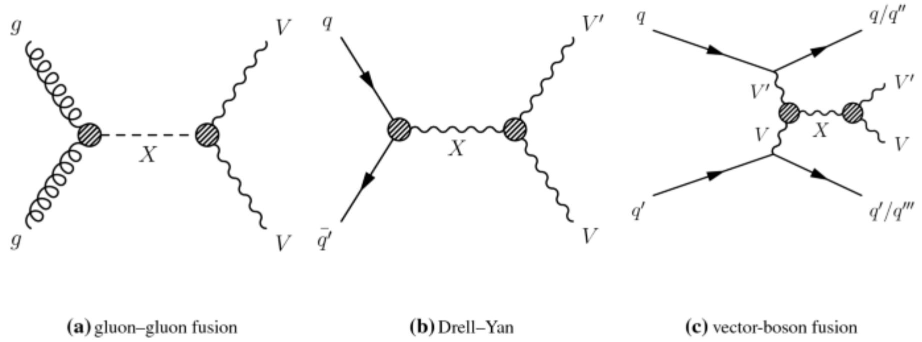


Figure 3: Feynman diagrams for the production of vector boson pairs [13]

2.2 Effective Field Theory and CP Violation

EFT is an effective theoretical framework used to model physics at low energy scales with respect to some energy scale Λ [25]. Effective field theories are typically characterized by light ($m \ll \Lambda$) and heavy ($M \gg \Lambda$), or low-energy and high-energy, degrees of freedom [16, 25]. Effective field theories are described by Lagrangians formed by the sum of operators. The SM characterizes the Lagrangian up to dimension four [26]. Any observable involving light degrees of freedom can be described by a Taylor expansion of the SMEFT in powers of $1/\Lambda^n$ [16]. Therefore, any expansion to the SM can be encoded in the SMEFT through implementation of higher-dimensional operator terms [16]. Any local interaction in the SMEFT can be described by:

$$\mathcal{L}_{\mathrm{EFT}} = \mathcal{L}_{\mathrm{SM}} + \sum_i \frac{c_i}{\Lambda^{d_i-4}}\mathcal{O}_i \tag{2}$$

where $\mathcal{O}_i$ are operators of mass dimension i constructed with the light fields, Λ is scale of heavy fields outside the Standard Model, and c_i represents the respective Wilson Coefficient [16, 25, 26].

Due to the exponential suppression of higher dimensional terms, typically only BSM operators up to dimension six are considered [26, 27]. Several of the dimension six operators described in Ref. [27] introduce CP-odd interactions between gauge bosons [18, 19, 21, 22, 23]. The squared matrix element for these EFT can be described by:

$$|\mathcal{M}|^2 = |\mathcal{M}_{\mathrm{SM}}|^2 + c \cdot 2Re(\mathcal{M}^*_{\mathrm{SM}}\mathcal{M}_{\mathrm{CP-Odd}}) + c^2|\mathcal{M}_{\mathrm{CP-Odd}}|^2 \tag{3}$$

where c represents the CP-odd operators' Wilson Coefficients [18, 19]. The first and third terms of Eq. 3 are CP-even, while the second term is CP-odd. For the remainder of the report, the second term will be referred to as the interference term.

Observable CP violating effects are generated by the interference between different amplitudes contributing to the same physical transition [15]; Eq. 3 demonstrates this as the interference between the CP-even SM amplitude $\mathcal{M}_{\mathrm{SM}}$ and the CP-odd BSM amplitude $\mathcal{M}_{CP-odd}$. The interference term changes sign under a CP transformation and therefore creates an asymmetric contribution to any CP-odd observable. In contrast, the squared terms of Eq. 3 are CP-even and symmetric about zero. Therefore, any measurable CP-violating asymmetry will only manifest in the interference term, motivating the construction of CP-odd observables sensitive to its sign. Bahl *et al.* proposes the discriminant $(P(+) - P(-))$ as an observable for examining this interference term [18]; where $P(+)$ $[P(-)]$ represents the probability for an event to correspond to positive [negative] interference.

In this work, EFT operators that modulate the amplitude of the $W^* \to WZ$ diboson production in Eq. 1 are considered. The corresponding Wilson coefficients are c_W for the BSM CP-even operator, and $c_{\tilde{W}}$ for the BSM CP-odd operator. Both Wilson coefficients have a value of +5 in the simulation.

2.3 Jet Reconstruction and Observables

Due to the high-energy and short lifetime in particle physics, many processes involving hadrons are modelled by jets: collimated sprays of particles that reveal information about the primary particle that created the jet. The process of identifying which particle created a jet is known as jet-tagging [28]. A common algorithm for reconstructing jets is the anti-k_t algorithm, which improves resilience of jet boundaries to soft-radiation, the clustering of low-energy particles around high energy particles [29]. Engineered observables that describe the collective characteristics of the jet are called high-level observables, while those of individual particles within the jet are called low-level constituent observables. In this work, semileptonic diboson decays ($VV \to \ell\ell qq$) are considered, where one boson decays hadronically and a large-radius jet is used to reconstruct the decay.

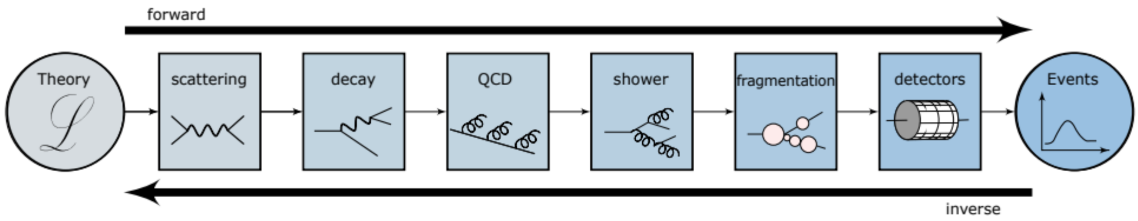


Figure 4: Diagram demonstrating processes of decay, jet formation, and detection in particle collider experiments [28].

2.4 Machine Learning

With an ongoing increase in available computational power, machine learning has become an increasingly accessible method of data analysis. Due to its ability to detect patterns and features in large datasets, machine learning is an important tool for particle physics, where data can be collected on the order of $1Pb$ per second [28].

A common use of machine learning is in classification tasks, where output data are discrete values corresponding to a given class. The objective of a machine learning model in this case is to calculate the probability of a given data point being in each of the possible classes. For a dataset with two potential classes, this task is called binary classification.

Neural networks are a method of machine learning inspired by early models of how neurons work within the human brain [30]. Neural networks consist of an input layer containing the feature values for each data point in the input set, an output layer containing the corresponding label for the given data point, and one or more hidden layers which progressively learn more complex representations of the data to map the inputs to the outputs. Figure 5 demonstrates the interaction between the input layer and first hidden layer [30]. Each computational unit within a layer is called a ’neuron’ or ’node’.

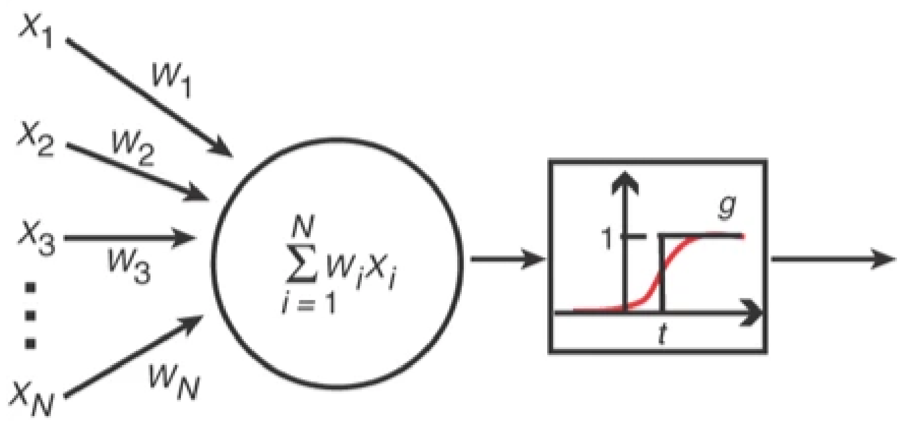


Figure 5: Diagram demonstrating the operation of a single neuron [30]. Inputs X_i are multiplied by corresponding weights W_i and summed together. The result of the linear sum is then fed through an activation function $g(\sum W_i X_i)$. This non-linear activation allows the network to learn complex relationships beyond a simple linear sum.

Mimicking the brain's sensory processing method, each neuron in one layer is used as an input for each neuron in the next layer. For example, each input feature value X_i will be fed to the first neuron of the hidden layer $h_1^{(2)}$ where the list of weights corresponding to this neuron are used to calculate a linear sum $\sum W_i X_i$. This resulting sum is then fed to an activation function to access complex non-linear relationships in the data. The output value of this neuron is then used as an input for the next layer of the network until the final output is processed.

The weights of each layer are initialized as random values. The learning algorithm iteratively updates these weights through a process known as gradient descent, in which the weights tend towards a minimum of the defined loss function, a measure of the model's performance. In a neural network, this process is completed using the backpropagation method, where the gradients and error in each layer are determined by progressing backward through the model architecture [31].

In this work, two distinct types of neural network were used: Deep Neural Networks (DNN) and Graph Neural Networks (GNN). DNNs are a type of feed-forward network, in which input data is fed forward through progressing weights and hidden layers until reaching the output layer; DNNs are typically characterized by having 3+ hidden layers, with smaller models being named shallow neural networks [32].

Graph Neural Networks are an extension of previous neural network models that can interpret data described by a graph, featuring nodes or feature values, and edges, a description of connection between nodes. Typically, tasks using GNNs are divided into two categories: graph-focused, in which the model will use the entire graph to predict and is not strictly dependent on individual nodes, or node-focused where the output depends on the properties of each node [33]. Common applications of a graph neural network include node classification, link prediction, community detection, node regression and edge regression, and graph classification and graph regression [34]. Figure 6 provides a simple demonstration of how a GNN may process data [35].

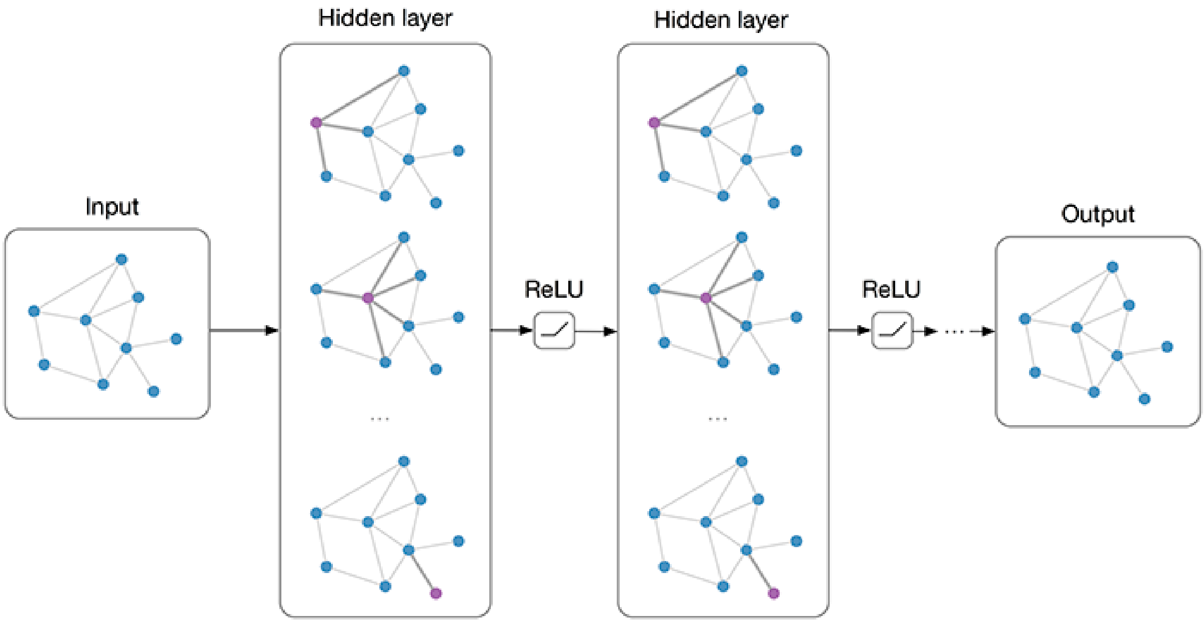


Figure 6: A demo model of a Convolutional Graph Neural Network [35]

GNNs provide an advantage in particle physics due to their invariance to permutations in the data [36]. Therefore, unlike DNNs which required fixed-size, ordered input, GNNs are able to analyse the unordered, variable length low-level constituent information provided from particle jets. This allows for the investigation into CP violation effects without the dimension reduction required to construct high-level observables, providing an opportunity to probe deeper into the process. Additionally, as GNNs incorporate edge information into their learning, geometric interactions between constituent particles are taken into consideration without additional feature engineering.

In this work, a DNN and GNN are used for the binary classification task of predicting the sign of the interference term. A machine-learning based observable, the discriminant $(P(+) - P(-))$, is then constructed to create the describe the production amplitude distributions. Derivations from the SM-predicted distributions can be used to identify the presence of CP-odd BSM operators.

3 Methods

3.1 Datasets and Simulation

The datasets used in this analysis were generated from simulated proton-proton collisions with a centre-of-mass energy of $\sqrt{s} = 14$ TeV, corresponding to the expected conditions of LHC Run 4 [37]. The detector effects used in this were modelled using the `Delphes` framework [38]. This creates a novel search into CP violating effects in diboson production at high-energy scales.

Input datasets can be grouped into two categories: high-level observables and low-level constituent observables. High-level observables represent summary statistics of the particle jet. Examples of high-level observables include the energy of the large-R jet, momentum of the leading subjet, and momentum of the positive lepton. The high-level dataset contained 32 training features and 1 label feature. The *Event Weight* feature was removed from training, however it was used to weight histogram outputs during post-training asymmetry evaluation. A complete list of input features can be found in Appendices A.1 and A.4.

Table 1: Summary of dataset composition and preprocessing.

Dataset type	Observables per event	Used for model
High-level (HL)	32 reconstructed kinematic variables	DNN
Low-level (LL)	8 low-level observables per constituent particle	GNN
Label feature	sign(*Event Weight*)	Binary target

To avoid data leakage between training and evaluation, the data were partitioned by event number parity; even-number events were used for training and validation, while odd-number events were reserved for testing. This provides 272773 training events and 273656 testing events. A separate model was trained on odd-number events and tested on even-number events. The models' training and evaluation were independent except in the production of final analysis plots which used the combined testing results of both models and datasets.

Constituent observables represent the statistics of the constituent particles forming the jet. For example, low-level observables, among other features, include the angular components, charge, and energy of each constituent particle. These observables may provide an advantage over the high level observables in searching for CP violation due to the latter's information loss when reducing dimension and the former's ability to maintain geometrical relations between constituent particles.

A separate label feature was created using the sign of *Event Weight* (1 for *Event Weight* > 0, and -1 for *Event Weight* < 0). The same label feature was used for the DNN and GNN. The task assigned to each model was therefore a binary classification task to predict the sign of the interference term, i.e. whether interference is constructive of destructive.

Four datasets are used for training and/or evaluation in this work. Each model was trained on a linear term dataset corresponding to the interference term in Eq. 3. The model was trained to predict whether interference was constructive or destructive as the sign of the interference term changes under a CP transformation, making it a CP-odd label. The CP-odd observable constructed from the model output should be asymmetric in the interference term, with deviation from asymmetry providing insight into the sensitivity to CP violations. As this is the only CP-odd term in Eq. 3, this is where information on CP-violating effects will manifest.

The trained models were also used to predict on two quadratic term dataset corresponding to CP-even and CP-odd operator contributions to $|\mathcal{M}_{\rm SM}|^2$ and $c^2|\mathcal{M}_{CP-odd}|^2$ in Eq. 3. Unlike the interference term, the quadratic contribution is CP-even and symmetric, providing a useful control sample. The final dataset is the SM dataset, which contains the SM term of Eq. 3 without additional BSM operators. A BSM CP-even term dataset was also examined.

3.2 Evaluation Metrics

Several evaluation metrics were used to evaluate the performance of the model; chosen metrics were consistent between DNN and GNN models to ensure fair comparison between their performance. Training loss is a measure of how much the model's predicted labels differ from true labels. Binary Cross Entropy with Logits Loss, `BCEWithLogitsLoss`, was chosen for this work. The loss function is mathematically described as:

$$\ell(x, y) = \frac{1}{N} \sum_{n=1}^{N} \ell_n, \qquad \ell_n = -\Big[y_n \log \sigma(x_n) + (1 - y_n) \log \big(1 - \sigma(x_n)\big) \Big], \tag{4}$$

where $\sigma(x_n) = \frac{1}{1+e^{-x_n}}$ is the sigmoid activation function, $y_n \in \{0, 1\}$ denotes the target label for the n-th event, and x_n is the predicted logit. By combining the loss and sigmoid activation into a single class, `BCEWithLogitsLoss` is more numerically stable than applying predicted probabilities directly into a binary cross entropy loss function [39].

To more intuitively evaluate the model's classification performance, Area Under the Receiver Operating Characteristic Curve (AUC score) was calculated using `scikit-learn` [40]. AUC score depicts the trade-off between true positives and false negatives of a classifier and is a commonly used metric for evaluating whether a model is detecting a signal [40, 41]. Additional evaluation metrics include precision (the ratio of true positives to total predicted positives), recall (the ratio of true positives to true positives and false negatives), and F1-score (the harmonic mean of precision and recall); these metrics were evaluated using `scikit-learn` [40]. The binary cross-entropy loss was monitored during training to optimise model weights, while the AUC score served as the primary evaluation metric for model comparison and hyperparameter tuning. Precision, recall, and F1-score provided additional insight into class balance and prediction reliability.

3.3 Feature Engineering and Preprocessing

Feature engineering is the process of selecting and transforming input variables used to train machine learning models. Several feature engineering processes were utilized in this research. To reduce the impact of one single feature on output, inputs were scaled using the standard scaler from the `scikit-learn` library [40]. Following the removal of uninformative features, 32 baseline features were considered for further inspection.

Furthermore, mutual information was used to rank the importance of a given feature on the model's prediction; this too was calculated using the `scikit-learn` library [40]. Mutual information is a model agnostic way to quantify the amount of information one variable provides about another [42]. The MI between input features and the output class therefore reflects each feature's relevance to the classification task. Mutual information is mathematically described as [42]:

$$I(X;Y) = \sum_{x \in X} \sum_{y \in Y} p(x,y) \log \frac{p(x,y)}{p(x)\,p(y)}, \tag{5}$$

where $p(x, y)$ is the joint probability distribution of (X, Y) and $p(x)$ and $p(y)$ are the corresponding marginal distributions. A higher value of $I(X;Y)$ indicates a stronger statistical dependency between the feature and the target class.

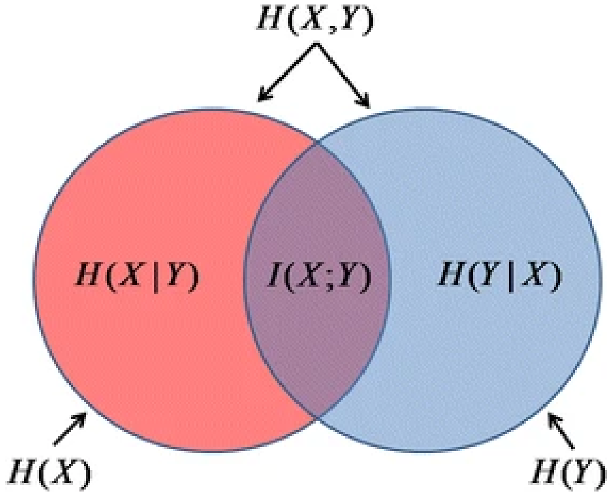


Figure 7: Diagram demonstrating the relationship between a variable's entropy, $H(X|Y)$, and mutual information, $I(X|Y)$ [42].

Figure 8 indicates the high-level observables most sensitive to CP-odd interference are the Leading Subjet Energy, the Subleading Subjet energy, and the azimuthal angles of the positive and negative leptons. Figure 9 demonstrates the distributions of the four high-level observables which provide visible separation between constructive and destructive interference. These distributions demonstrate clear separation between interference terms. Of the 32 baseline features, only 18 were found to provide non-zero mutual information on the interference term. Distributions for each high-level observable used in final training can be found in Appendix A.2.

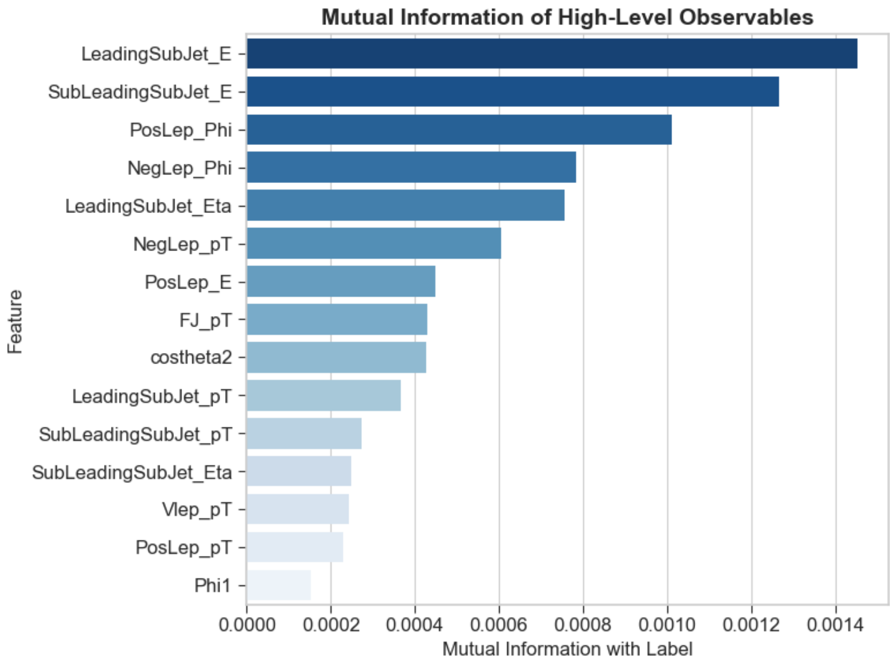
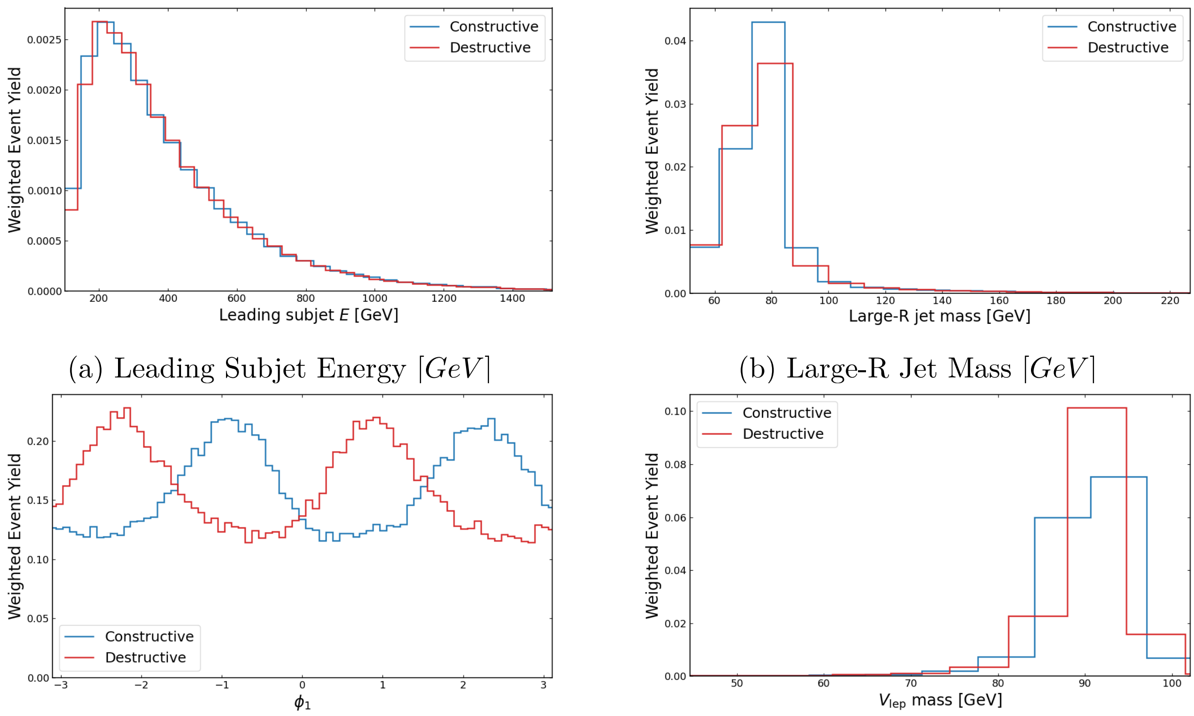


Figure 8: Graph demonstrating the mutual information provided by the top fifteen base high-level features. More informative features appear higher on the graph with a deeper shade, less informative features are placed lower with lighter shades.



(a) Leading Subjet Energy $[GeV]$

(b) Large-R Jet Mass $[GeV]$

(c) Azimuthal Angle

(d) Leptonic Boson Invariant Mass $[GeV]$

Figure 9: Distributions for the four high-level observables which provide the high mutual information. Blue indicates the distribution for constructive interference events, and red for destructive.

Polynomial feature engineering is the process of creating multiples of input features; for example, if a dataset has features X and Y, polynomial features could include X^2, Y^2, and, XY. In this work, the impact of high-level polynomial features on the DNN's performance was investigated. Performance of the DNN with polynomial features resulted in a decrease of AUC score on the order of magnitude $\mathcal{O}(10^{-2})$. Therefore, the final model was trained only including base features. Further details on the polynomial features' impacts can be found in Appendix A.5.

To further assess the impact of feature selection, the DNN was trained iteratively with progressively smaller subsets of high-level observables. For each iteration, the two observables with the lowest mutual information was removed. AUC scores and loss were calculated on a validation set for each model. Figure 10 demonstrates the AUC score and loss as a function of features included. The models discriminating power remained relatively constant for ≥ 18 features indicating that the most useful information is held in the top-ranked observables. Performance began to drop for ≤ 15 features, decaying towards random classification.

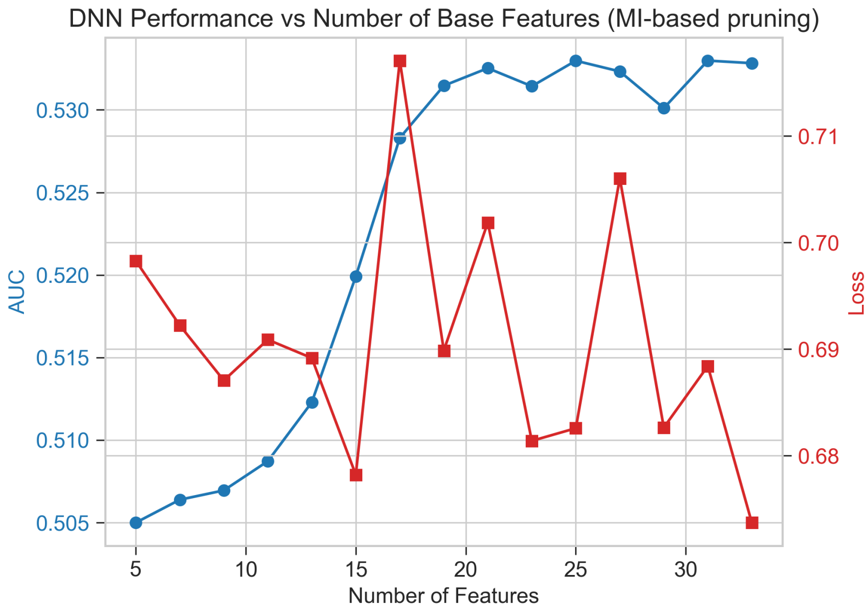


Figure 10: Diagram showing the validation AUC (blue) and cross entropy loss (red) of the model as a function of number of baseline features included. Features were selected by ranking of mutual information. Performance was relatively stable between 18 and 32 features and shows sharp decline around 15 features indicating that many features do not provide much information on the interference term (classification label).

As performance was consistent between 18 and 32 features, the model was trained for 10 epochs with 18, 20, 25, and 32 features on five different random seeds to ensure feature training stability. The validation AUC score and loss were calculated for each training instance and the mean and standard deviation of AUC was calculated for each subset of features. Figure 11 shows the results of this test; it is clear that the removal of all features with zero MI shows the best overall performance. Therefore, 18 features were used for training the final model. The 18 features used in training the DNN model are described in table 2. Details on every high-level observable can be found in Appendix A.1.

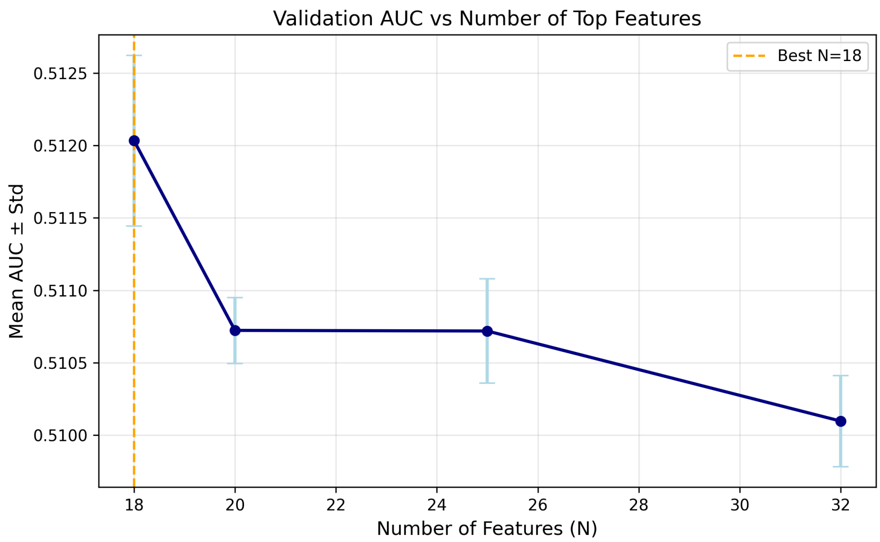


Figure 11: Diagram showing the mean validation AUC score over five separate random seeds as a function of number of features included. Error margins were calculated by standard deviation of AUC score. The highest performing subset was 18 features, indicating that all non-zero MI features are used in the model's thinking. Thus, the final model was trained used only non-zero MI features.

Table 2: High-level observables with non-zero mutual information (MI) relative to the target label, used as input features in DNN training. These observables demonstrate measurable dependence on the CP-sensitive classification target and thus provide the most informative kinematic descriptors.

Feature	**Mutual Information**	**Description**
LeadingSubJet_E	0.00145	Energy of the leading subjet
SubLeadingSubJet_E	0.00106	Energy of the subleading subjet
PosLep_Phi	0.00097	Azimuthal angle of the positively charged lepton
NegLep_Phi	0.00076	Azimuthal angle of the negatively charged lepton
LeadingSubJet_Eta	0.00075	Pseudorapidity of the leading subjet
NegLep_pT	0.00064	Transverse momentum of the negatively charged lepton
PosLep_E	0.00051	Energy of the positively charged lepton
SubLeadingSubJet_Eta	0.00048	Pseudorapidity of the subleading subjet
FJ_pT	0.00047	Transverse momentum of the large-R jet
LeadingSubJet_pT	0.00044	Transverse momentum of the leading subjet
costheta2	0.00035	Cosine of the decay angle of the second boson
PosLep_pT	0.00025	Transverse momentum of the positively charged lepton
Vlep_pT	0.00020	Transverse momentum of the reconstructed leptonic boson
Phi	0.00019	Azimuthal angle between leptonic and hadronic decay planes
FJ_eta	0.00011	Pseudorapidity of the large-R jet
Phi1	0.000095	Angular observable sensitive to CP-odd interference
SubLeadingSubJet_Phi	0.000083	Azimuthal angle of the subleading subjet
Vlep_eta	0.000051	Pseudorapidity of the reconstructed leptonic boson

Constituent data was preprocessed in data objects using the `PyTorch_Geometric` library [43]. Each collision event was processed into a single data object with constituent particle features representing the graph nodes. The graph edges were constructed using a k nearest neighbours approach for the angular features in the dataset, *Constituent_eta* and *Constituent_phi*. Therefore, the constituent particles were connected via angular separation; the constructed edges were made bidirectional. Figure 12 demonstrates an example data object used for training the GNN.

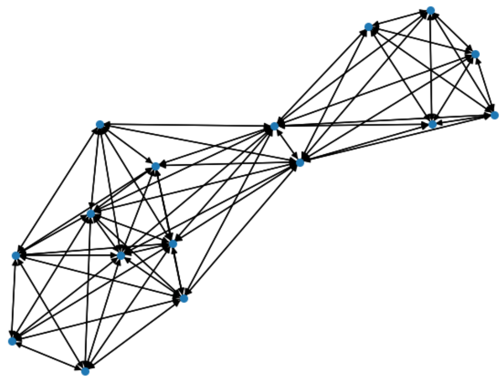


Figure 12: Diagram of example graph data object used for training of the GNN

During the GNN development, A separate method of feature selection was tested. That is a filter was applied such that only leptonic constituent particles were considered. This was used to emphasize the lepton final state in the $VV \to \ell\ell qq$ decay mode and reduce complexity of model training. The performance between a lepton-only trained model and a full-jet trained model was compared and it was found that the lepton-only model decayed to random classification (AUC $\to$ 0.5). Therefore, the full set of constituent particles was used for training the graph network.

3.4 Model Architectures

As described in section 2.4, two separate machine learning models were tested in this work: a feed-forward deep neural network (DNN) and a graph neural network (GNN). As described in section 3.1, the DNN was trained on high-level observables while the GNN was trained on constituent components. Therefore, each model has a distinct architecture independent of the other.

3.4.1 Deep Neural Network

A feed-forward Deep Neural Network (DNN) was implemented using the `PyTorch` framework to classify CP-odd and CP-even interference terms based on high-level observables [39]. The model used the high-level observables described in section 3.1 following recursive feature elimination described in section 3.3. Due to their reduction in performance, polynomial features were not considered in training.

The architecture for the final DNN model consisted of four hidden layers with dimensions $512, 256, 128,$ and 64 respectively. Each linear layer was followed by a batch normalization layer, which has been demonstrated to improve computational efficiency and model performance [44]. After batch normalization, Rectified Linear Unit, ReLU, activation functions were used for nonlinearity due to their simplicity and stability in deep networks. Dropout layers were applied with a dropout rate of 0.2 to minimize the chance of overfitting to noise by randomly turning off neurons during training. The DNN model architecture is represented in Figure 13.

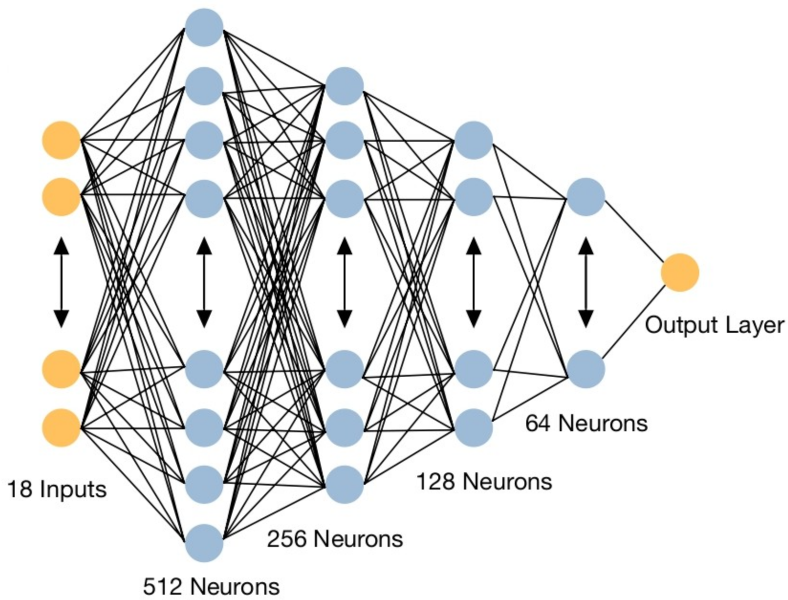


Figure 13: Diagram of the DNN architecture used. The model contains an input layer of size 18 followed by four ReLU hidden layers of dimension 512, 256, 128, and 64 respectively.

Two independent early stopping classes were created to prevent overfitting during training. The first evaluated the loss on validation data while the second evaluated AUC. The loss-based early stopper ensured validation loss continued to decrease beyond a minimum threshold, Δ_{loss}, while the AUC-based early stopper ensured validation AUC continued to increase above a minimum threshold, Δ_{AUC}. Both early stoppers had a patience of five epochs without validation performance increase before the training stopped. Furthermore, both values of Δ were set to 10^{-4}.

Hyperparameter optimisation was completed using the Bayesian optimisation method known as Tree-Structured Parzen Estimator (TPE) [45]. This was implemented via the `Optuna` library [46]. Table 3 demonstrates the details of the `Optuna` study including hyperparameter search space and determined optimal values. The input dataset was split in an 80/20 training/validation partition. Model performance was compared via validation AUC score. A total of 30 trials were completed with each trial completing 15 training epochs. The hyperparameters that resulted in the highest validation AUC score (0.5197) were used for the final model.

Table 3: Summary of hyperparameter optimisation for the Deep Neural Network (DNN) classifier performed using the `Optuna` framework. Each trial trained for 15 epochs and was evaluated by validation ROC–AUC score. The search space was explored using the TPE sampler with median pruning.

Hyperparameter	**Search Space**	**Best Value (AUC = 0.5197)**
Hidden dimension (h)	{64, 128, 256, 512}	512
Number of hidden layers	[2, 8] (integer)	4
Learning rate (η)	Log-uniform $[10^{-5}, 10^{-2}]$	1.7×10^{-4}
Batch size	{64, 128, 256, 512}	128
Dropout rate (p)	Uniform $[0.0, 0.5]$	0.20
Activation function	{ReLU, LeakyReLU}	ReLU
Optimizer	Adam (Fixed)	Adam
Loss function	Binary Cross-Entropy with Logits Loss (Fixed)	–
Epochs per trial	15 (Fixed for Optuna Study, else 20)	–
Training / Validation split	80 / 20 (Fixed for Optuna study, else event-number parity split)	–
Search strategy	TPE sampler (seed = 42), median pruning after 5 trials	
Number of trials	30 trials	
Objective metric	Validation ROC–AUC (maximize)	

The DNN provides a baseline model that is exclusively trained on high-level observables. In contrast, the subsequent Graph Neural Network in section 3.4.2 leverages low-level constituent observables to capture geometric correlations that are lost during feature reconstruction.

3.4.2 Graph Neural Network

Three different GNN frameworks were tested in this work: A simple Graph Convolutional Network (GCN), a physics-inspired GCN based on the ParticleNet model which excels in Jet Tagging classification tasks [47], and a graph implementation of ResNet which carries a residual of the input layer passed hidden layer outputs [48]. The three models were put through the training pipeline and relative performance was evaluated using AUC score. None of the three tested GNN models were able to generate meaningful results and thus GNN was not processed any further. The failure of the GNN likely occurred due to the extremely subtle CP structure at constituent level and the corresponding mismatch of model architecture. Further details describing the GNN used in this work can be found in appendix B.2.

3.4.3 Training Configuration

The DNN was trained for 20 epochs using an event-number parity based splitting approach. For each network, a model was trained on even event number events and tested on odd event number events. A second model was separately trained on odd event number events and tested on even event number events. The models trained on different event number parities remained independent and were only combined for plotting a discriminant histogram over the full dataset for physical analysis.

Following hyperparameter tuning, the DNN used a learning rate of 0.00017. Mini-batches of size 128 were processed using the ADAM optimizer [49]. Early stopping was initially applied when validation loss or AUC failed to improve by 10^{-4} for five consecutive epochs;

however, due to the subtle and volatile nature of model performance, this feature was removed from final training and the number of epochs was manually tuned.

The trained models were trained using `BCEWithLogitsLoss` to optimize model weights. The output label, the sign of *Event Weight*, represented whether the event had constructive or destructive interference between Standard Model physics and Beyond Standard Model physics with regards to CP-odd operators. The output of the model was the probability that the interference term was positive, i.e. the probability interference was constructive. For each predicted output, the discriminant $(P(+) - P(-))$ was calculated.

3.5 Model Output Analysis

The model output is plotted in a histogram weighted by the corresponding *Event Weight* for the given event. This constructs a CP-odd observable, $D = P(+) - P(-)$, that can be visualized via a weighted histogram and describes the model's sensitivity to CP-violating effects. The distribution of the discriminant is expected to be asymmetric, or CP-odd, for the interference term, and symmetric for the other SM and quadratic terms as they are CP-even. The symmetry of quadratic terms acts as a control sample, verifying the model is not fabricating artificial asymmetries. As the interference term magnitude is linear, it is expected to be much smaller than the quadratic terms. This supports the expectation of CP-violating effects being subtle in diboson production, and that strong separation between distributions is not expected.

3.6 Additional Physics Observables

Similarly, distributions for the other high-level observables are plotted for the linear interference term, quadratic term, and SM term datasets. While the model was only trained on high-level observables listed in table 2, the full set of 32-high level observables were plotted for the purpose of interpreting the behaviour of the SMEFT contributions. The variations in the distributions between datasets reveals insights into the observables' sensitivities to BSM effects and CP-structure. Additionally, a composite feature, the VV system invariant mass, was constructed with:

$$M_{WZ}^2 = m_W^2 + m_Z^2 + 2(E_W E_Z - \vec{p_W} \cdot \vec{p_Z}) \tag{6}$$

where m_W, E_W, and $\vec{p_W}$ represent the invariant mass, energy, and three-momentum of the W boson and m_Z, E_Z, and $\vec{p_Z}$ represent the invariant mass, energy, and three-momentum of the Z boson respectively. This feature was not provided to the network during training, but it is included in analysis as SMEFT contribution scale with energy and therefore produce characteristic enhancements at high invariant-mass levels.

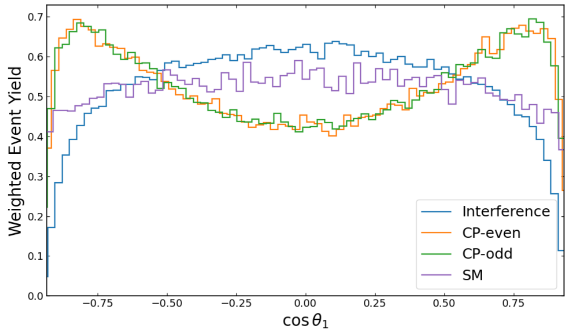
4 Results & Discussion

This section presents the distribution of high-level observables in the interference term, quadratic terms, and SM term of Eq. 3. Analysis of these distributions provides insights onto an observables sensitivity to CP-structure and BSM effects. Additionally, this section presents the performance of the trained neural networks and investigates their ability to construct a CP-odd observable to probe CP-odd interference. The results are analysed in terms of model discriminants, asymmetry observables, and comparison to SM and quadratic BSM control samples.

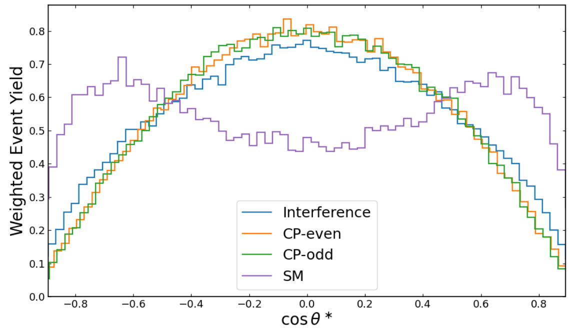
4.1 High-Level Observable Sensitivity to SM and BSM effects

To examine the sensitivity of high-level observables to SM and BSM effects, each high level observable's distribution in the four matrix element terms were plotted as histograms. The distributions of six high-level observables are displayed in this section. These observables are grouped into two groups: angular observables, which can be sensitive to CP-structure, and energy sensitive observables, which are sensitive to BSM effects. The distributions of all 32 high-level observables can be found in Appendix A.3.

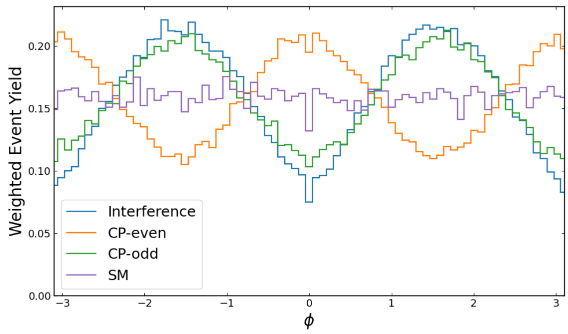
Figure 14 shows the distributions of three selected angular observables: $\cos\theta_1$, $\cos\theta^*$, and ϕ. Figure 14a, the distribution for $\cos\theta_1$, shows moderate separation between SM term distributions and BSM distribution, indicating the 1st boson decay angle is particularly sensitive to BSM physics. In contrast, the distribution for $\cos\theta^*$ in Figure 14b shows minimal separation between SM and BSM terms, but it demonstrates a visible separation between interference terms. Like $\cos\theta^*$, the distribution for ϕ in Figure 14c shows a visible separation between BSM and SM terms. Additionally, the distribution visibly contains a moderate separation between CP-even and CP-odd quadratic terms. This suggests that ϕ is sensitive to CP-structure within the BSM term of the SMEFT in addition to BSM enhancements. In summary, $\cos\theta_1$ is sensitive to BSM effects, $\cos\theta^*$ is sensitive to the interference term between SM and CP-odd BSM effects, and ϕ is sensitive to BSM effects and their CP-structure.



(a) $\cos\theta_1$



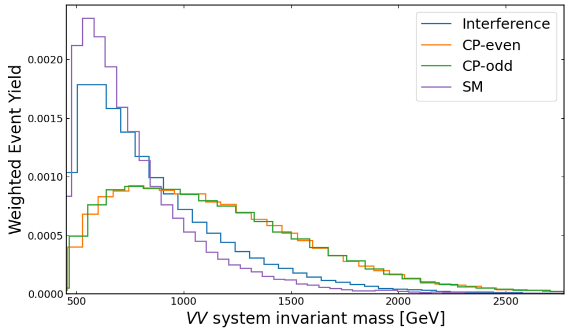
(b) $\cos\theta^*$



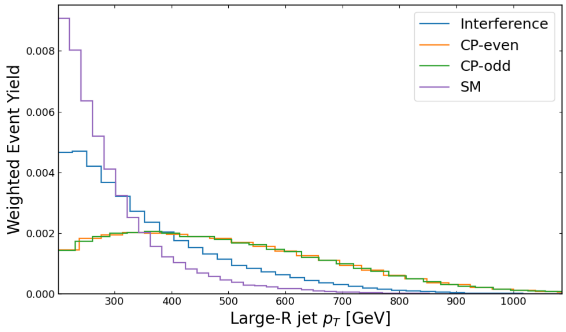
(c) ϕ

Figure 14: Distributions of selected high-level angular observables ($\cos\theta_1$, $\cos\theta^*$, and ϕ) for the SM, CP-even, CP-odd, and interference matrix-element terms. Angular variables probe the decay geometry of the diboson system and exhibit distinct sensitivity to CP structure.

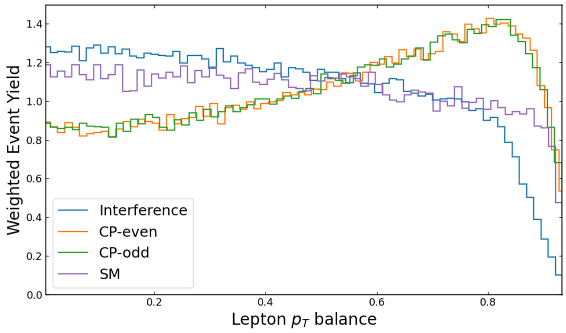
Figure 15 displays the distributions of three energy-dependent observables: VV invariant mass, Large-R jet transverse momentum, and lepton transverse momentum balance. As BSM terms alter the energy of the system, as demonstrated in Eq. 2, energy dependent observables can be particularly sensitive to BSM physics. Figure 15a demonstrates the distribution of the invariant mass of the diboson system; the figure demonstrates visible separation between SM and BSM terms, with BSM distributions reaching higher values of invariant mass as expected by SMEFT. Figure 15b demonstrates the distribution of the Large-R jet transverse momentum and similarly contains a moderate separation between BSM and SM terms with the former reaching larger values. Finally, the lepton transverse momentum balance distribution in Figure 15c shows a moderate separation between BSM and SM terms with the BSM peak have a higher momentum balance value than the SM peak. In summary, these three energy-dependent high-level observables demonstrate visible separation between SM and BSM terms, showing sensitivity to BSM effects. This is expected by SMEFT as the inclusion of BSM operators scales the energy dependence of the amplitudes.



(a) VV invariant mass [GeV]



(b) Large-R jet transverse momentum (p_T)



(c) Lepton transverse momentum balance

Figure 15: Energy-sensitive observables for the SM, CP-even, CP-odd, and interference matrix-element terms. The VV invariant mass and large-R jet p_T show strong separation between SM and BSM physics due to enhanced high-energy tails, while the lepton p_T balance probes recoil differences in diboson decay.

As demonstrated, the high-level observables examined in this work varying sensitivity to BSM effects, interference contributions, and in some cases to the CP-structure of the BSM SMEFT operators. Energy-based observables, such as the diboson system invariant mass demonstrated in Figure 15a, contain increased sensitivity to BSM effects as energy scales with the inclusion of BSM operators in SMEFT. Certain angular observables, such as ϕ demonstrated in Figure 14c, generally contain moderate separation between CP-even and CP-odd quadratic terms, showing sensitivity to CP-structure in BSM physics. A complete set of feature distributions can be found in Appendix A.3.

4.2 DNN Model Performance

The CP-violating effects only manifest within the interference term. As a result, even optimally trained classifiers cannot achieve strong separation between positive and negative interference labels. Modest performance values close to 0.50 are therefore expected and do no indicate poor model quality, but instead the subtlety of CP-violating physics in $WZ \to \ell\ell qq$ events. Information on the effect of CP violation in diboson production is extracted by using the machine-learning based observable $(P(+)-P(-))$, and comparing the measured distributions against SM and BSM hypotheses.

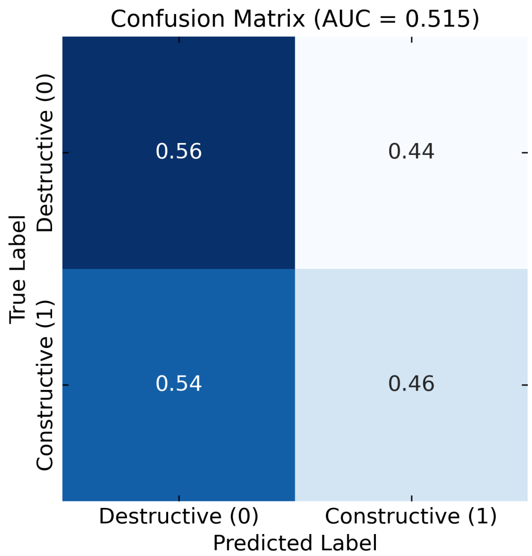
Table 4: Table comparing the performance metrics between GNN, DNN trained on even event numbers, and DNN trained on odd event numbers. Classification metrics include AUC score, precision, recall, and F1-score.

Model	Interference	**AUC**	**Precision**	**Recall**	**F1-score**
DNN (even events)	Constructive	0.515	0.51	0.46	0.48
	Destructive	–	0.51	0.56	0.54
DNN (odd events)	Constructive	0.514	0.51	0.52	0.51
	Destructive	–	0.51	0.50	0.50

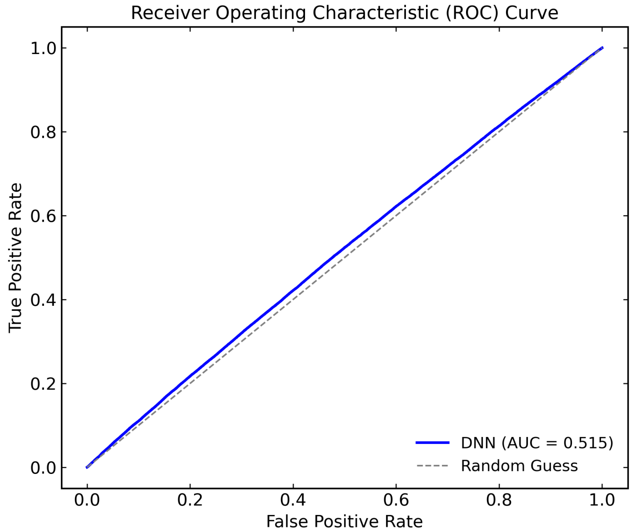
Table 4 compares the performance of the DNN models on the linear term dataset. The DNN model trained on even- and odd-indexed event numbers achieved AUC scores of 0.515 and 0.514 respectively. These metrics are modestly above random; however, these results are consistent with the subtlety of CP-violating effects.

Precision, recall, and F1-scores remain close to 0.5 for all models, highlighting the weak correlation between final-state observables and the sign of the interference term. The relatively similar performance between the two DNN models indicates training stability under the parity-based partitioning approach used in this work. These performance metrics are not primary indicators of CP-sensitivity, but instead serve to verify the models are able to learn weak discriminative patterns which manifest as asymmetric distributions, analysed in section 4.3.

Figure 16 contains the confusion matrix and ROC curve for the DNN model trained on even-indexed event numbers. Figure 16a indicates the DNN model is able to pick up weakly predictive patterns in the data; however, the model appears to slightly skew towards one class as evidenced by higher performance on predicting destructive interference in Fig. 16a. The ROC curve in Figure 16b further indicate the model learning weakly predictive patterns in the data. A detailed comparison of the confusion matrices and ROC curves between the even- and odd-indexed DNNs can be found in Appendix B.1.



(a) Confusion matrix for model trained on even-indexed event numbers. The model has a good true negative rate of 0.56, and a low false positive rate of 0.44. However, the other metrics are underperforming, highlighting the subtlety in connecting kinematic observables to CP-odd interference.



(b) ROC curve for model trained on even-indexed event numbers. The small area above guessing indicates only a weak correlation between the kinematic observables and sign of the interference term.

Figure 16: Confusion Matrix and ROC curve for DNN model trained on even-indexed event numbers. The graphs indicate a slight ability for the model to learn predictive patterns within the high-level observables dataset.

4.3 Discriminant Based CP-Odd Observable

The trained DNN enables construction of a CP-odd observable, $D = P(+) - P(-)$, whose distribution directly reflects the sign structure of the interference term. Asymmetry in the interference term arises under CP-transformation; the asymmetry within the interference term in Figure 17 confirms the constructed observable is CP-odd. SM and quadratic terms are CP-even and thus are expected to be symmetric; Figure 17 further demonstrates symmetry within SM and quadratic BSM terms, verifying the model is not fabricating asymmetry.

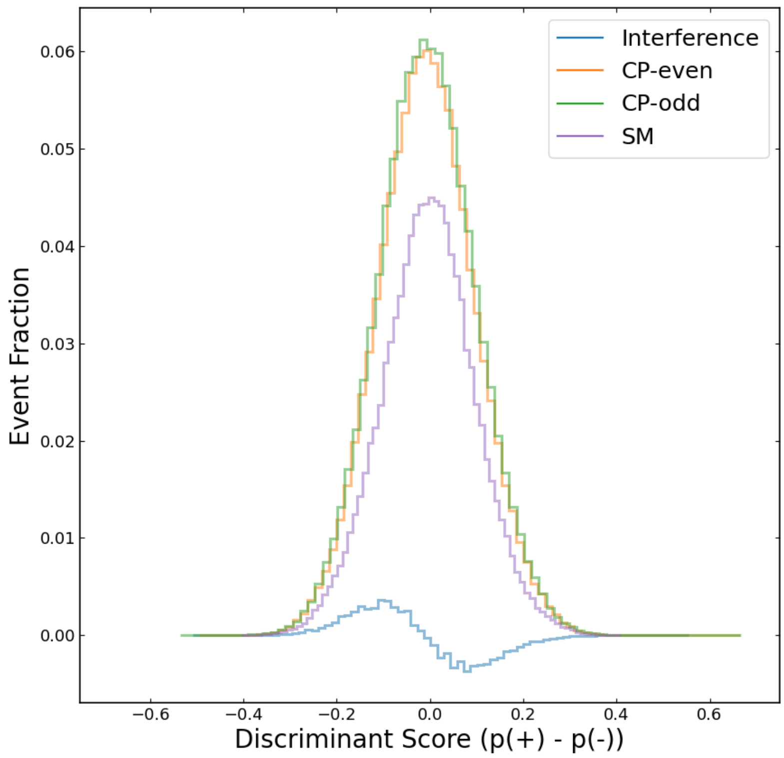


Figure 17: Histogram of the machine learning based observable, $P(+) - P(-)$, for the CP-even and CP-odd quadratic terms, SM term, and the linear interference term. Distribution shows expected asymmetry for interference term and symmetry for quadratic and SM terms.

As the magnitude of the interference term scales with c, compared to c^2 for quadratic terms in Eq. 3, its contribution to overall cross section is suppressed. This is verified in Figure 17 as the scale of the interference term is dwarfed by the scale of the quadratic terms. As the only CP-odd term within Eq. 3 is the interference term, the relatively small magnitude of interference accounts for the subtlety of CP-violating effects. This provides explanation for the low classification metrics of the DNN described in section 4.2. As the primary evidence of CP-sensitivity is the distribution rather than performance metrics, the DNN model has successfully enabled the creation of a machine-learning based CP-odd observable.

5 Conclusion & Outlook

This work demonstrated the ability to construct a machine-learning based CP-odd observable sensitive to CP violation in the SMEFT interference term. A deep neural network was trained on high-level observables to predict whether interference between SM and

BSM terms was constructive or destructive. The model output was used to construct the CP-odd observable: $P(+) - P(-)$. A graph neural network was also tested; however, it was unable to learn any meaningful patterns in the low-level constituent data; the graph network failure likely occurred due to the extremely subtle CP structure at constituent level and the corresponding mismatch of model architecture.

The distribution of the discriminant displayed the expected asymmetry within the interference term, confirming the observable is CP-odd. Additionally, symmetry in SM and BSM quadratic terms confirms the model is not fabricating asymmetries that were not present. The distributions of high-level observables also revealed varying sensitivity to BSM effects, interference contributions, and in some cases to the CP-structure of BSM terms.

Overall, this work demonstrates that a machine-learning based observable can successfully reproduce the CP-odd structure of the interference term in diboson production. The constructed discriminant provides a new proxy for CP violation in collider physics and highlights the potential of simulation-based inference methods in probing subtle effects beyond the Standard Model. Future work could include developing the model for additional final states. As constituent data maintains the geometric correlation within the data, the development of a graph neural network remains an area of interest and more state-of-the-art models are recommended for testing.

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All code used for data processing, feature engineering, model training, and figure generation is made publicly available at:

`https://github.com/colelooney/SH-Project`

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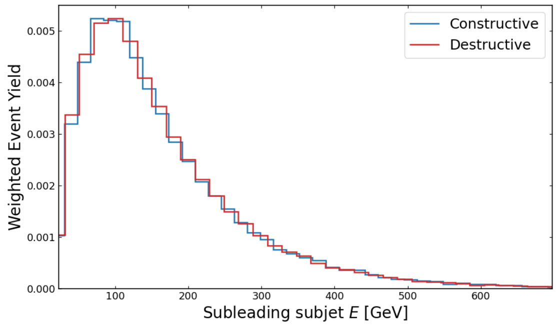
A Dataset Features

A.1 High-Level Observables

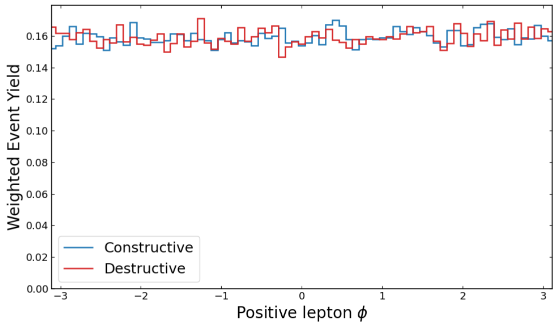
Table 5: Summary of high-level observables used as input features for the Deep Neural Network (DNN). These observables represent reconstructed quantities derived from detector-level data, corresponding to global event and jet-level kinematics.

Feature	**Description**
FJ_E	Energy of the reconstructed large-R jet
FJ_eta	Pseudorapidity (η) of the large-R jet
FJ_flavour	Jet flavour label (removed due to lack of variation)
FJ_mass	Reconstructed invariant mass of the large-R jet
FJ_pT	Transverse momentum (p_T) of the large-R jet
FJ_phi	Azimuthal angle (ϕ) of the large-R jet
LeadingSubJet_E	Energy of the leading subjet
LeadingSubJet_Eta	Pseudorapidity of the leading subjet
LeadingSubJet_Phi	Azimuthal angle of the leading subjet
LeadingSubJet_pT	Transverse momentum of the leading subjet
Lep_pT_balance	Ratio of positive to negative lepton transverse momentum
Event Weight	Event generator weight (used for weighting, not training)
NegLep_E	Energy of the negatively charged lepton
NegLep_Eta	Pseudorapidity of the negatively charged lepton
NegLep_Phi	Azimuthal angle of the negatively charged lepton
NegLep_pT	Transverse momentum of the negatively charged lepton
Phi	Azimuthal angle between leptonic and hadronic decay planes
Phi1	Angular observable sensitive to CP-odd interference
PosLep_E	Energy of the positively charged lepton
PosLep_Eta	Pseudorapidity of the positively charged lepton
PosLep_Phi	Azimuthal angle of the positively charged lepton
PosLep_pT	Transverse momentum of the positively charged lepton
SubLeadingSubJet_E	Energy of the subleading subjet
SubLeadingSubJet_Eta	Pseudorapidity of the subleading subjet
SubLeadingSubJet_Phi	Azimuthal angle of the subleading subjet
SubLeadingSubJet_pT	Transverse momentum of the subleading subjet
Type	Event type identifier
Vlep_E	Energy of the reconstructed leptonic boson
Vlep_eta	Pseudorapidity of the reconstructed leptonic boson
Vlep_mass	Invariant mass of the reconstructed leptonic boson
Vlep_pT	Transverse momentum of the reconstructed leptonic boson
Vlep_phi	Azimuthal angle of the reconstructed leptonic boson
cosThetaStar	Cosine of the helicity angle between bosons in the center-of-mass frame
costheta1	Cosine of the decay angle of the first boson
costheta2	Cosine of the decay angle of the second boson

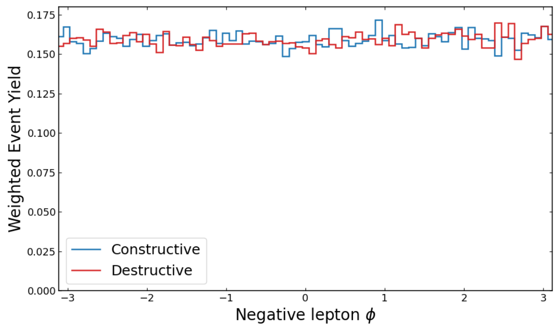
A.2 High-Level Observable Interference Distributions



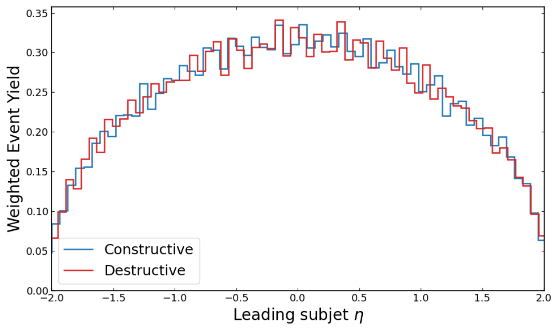
(a) Subleading Subjet Energy [GeV]



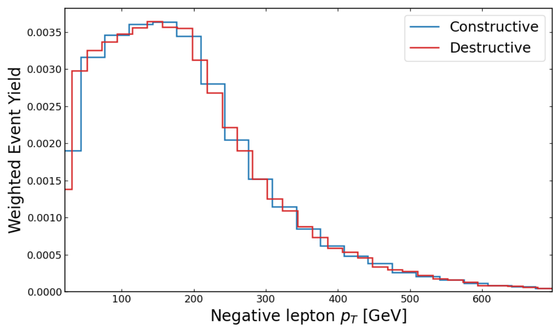
(b) Positive Lepton Azimuthal Angle



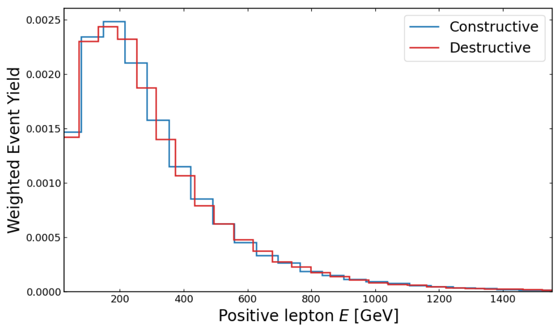
(c) Negative Lepton Azimuthal Angle



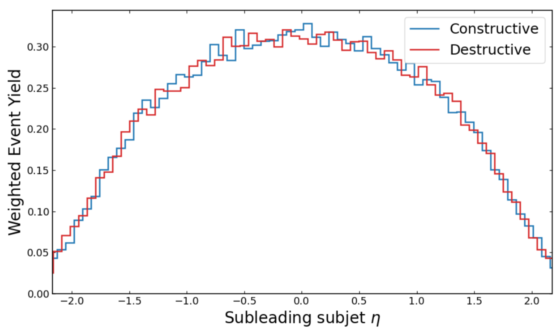
(d) Leading Subjet Pseudorapidity



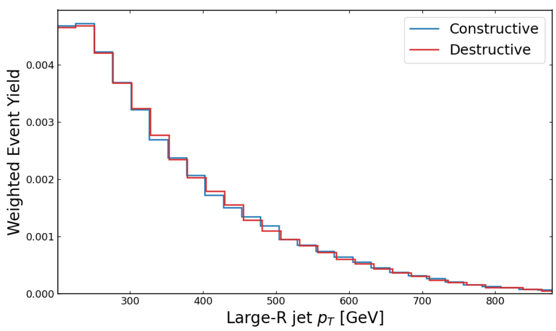
(e) Negative Lepton Transverse Momentum



(f) Positive Lepton Energy [GeV]

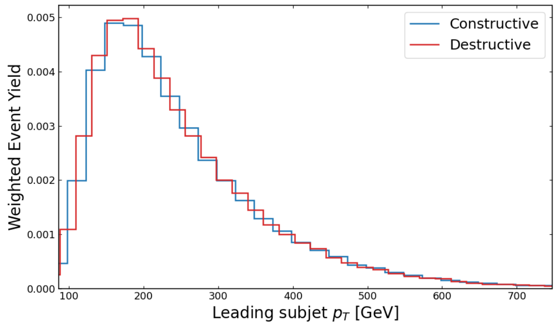


(g) Subleading Subjet Pseudorapidity

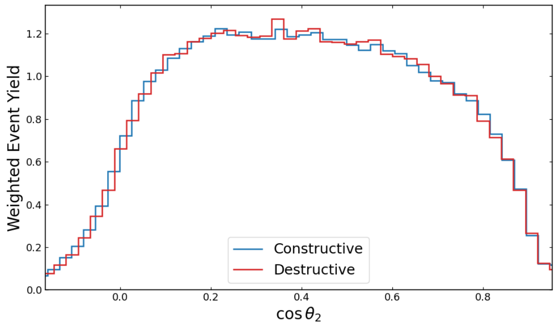


(h) Large-R Jet Transverse Momentum [GeV]

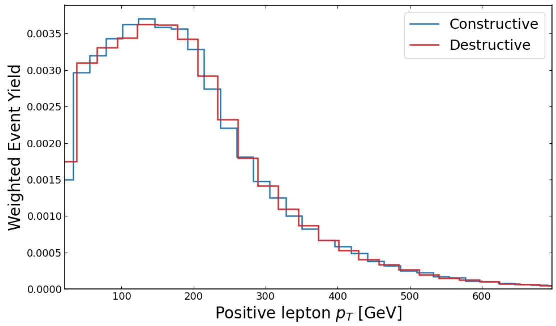
Figure 18: Distributions of high-level observables (Part 1) ordered by decreasing mutual information (MI).



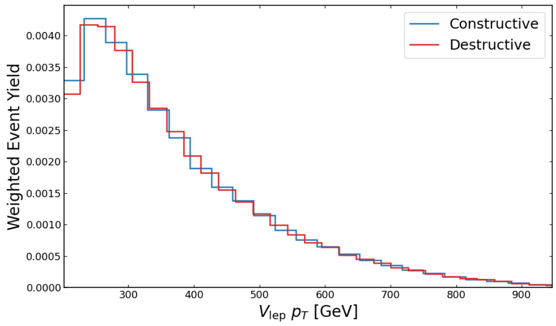
(a) Leading Subjet Transverse Momentum [GeV]



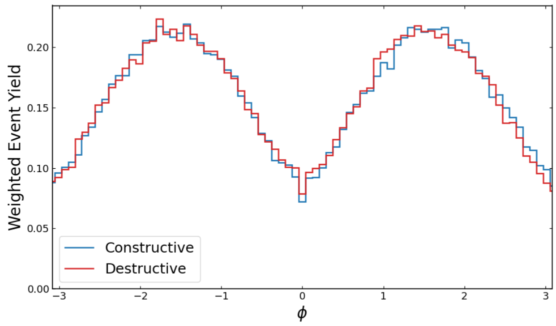
(b) Decay Angle of Second Boson ($\cos\theta_2$)



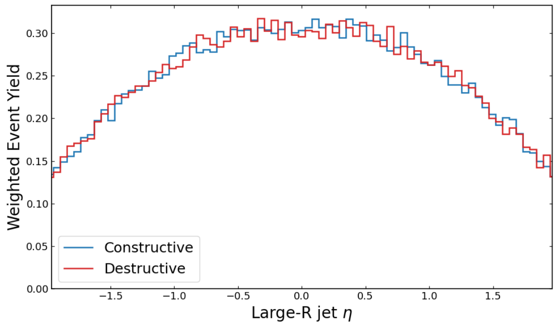
(c) Positive Lepton Transverse Momentum



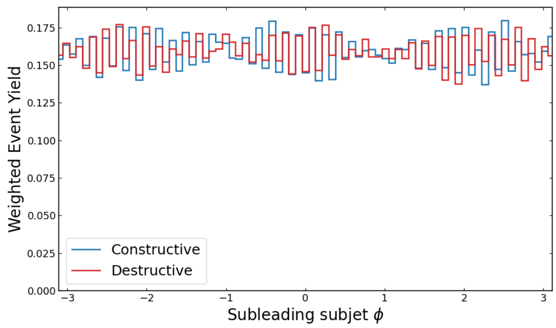
(d) Leptonic Boson Transverse Momentum



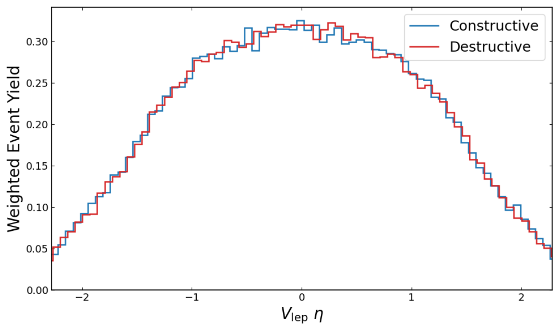
(e) Azimuthal Angle Between Decay Planes (ϕ)



(f) Large-R Jet Pseudorapidity



(g) Subleading Subjet Azimuthal Angle



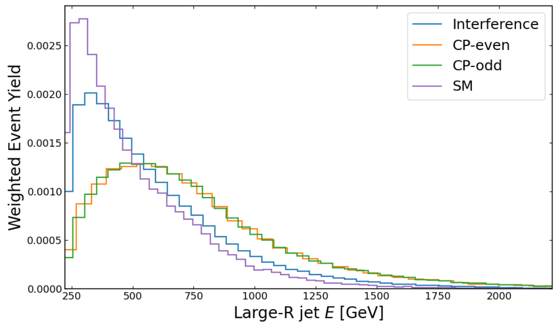
(h) Leptonic Boson Pseudorapidity

Figure 19: Distributions of high-level observables (Part 2) ordered by decreasing mutual information (MI).

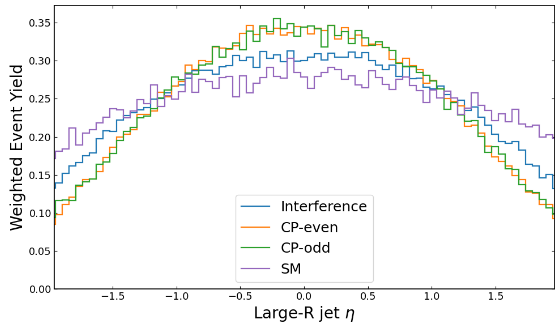
A.3 High-Level Observable SMEFT Distributions

The following figures show the distributions of all high-level observables used or considered in this analysis. Each observable is plotted for the four matrix-element components: the SM term, the CP-even quadratic term, the CP-odd quadratic term, and the linear interference term. All histograms are weighted by the corresponding *Event Weight*. Observables are grouped by their physical category for clarity. Many subjet features display singular peaks in the distribution, this is due to the feature having default values when a subleading subjet is absent.

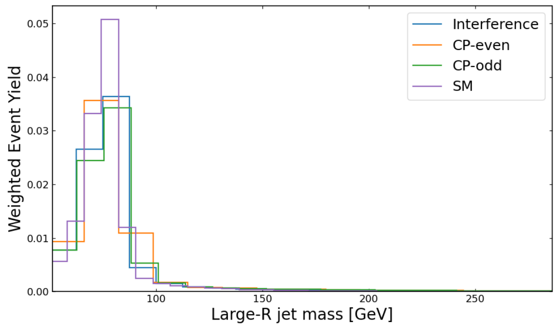
Large-R Jet Observables



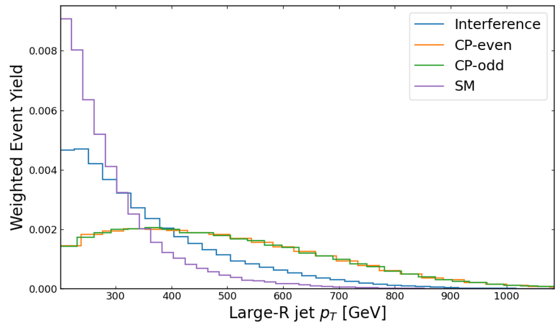
(a) Large-R Jet Energy



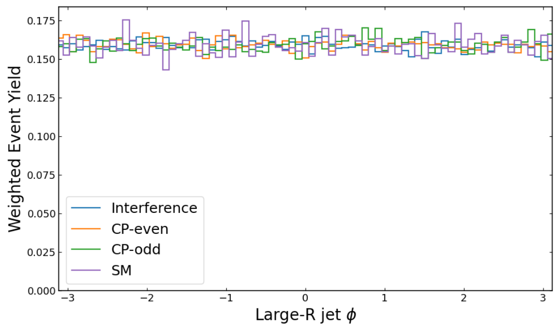
(b) Large-R Jet Pseudorapidity (η)



(c) Large-R Jet Mass



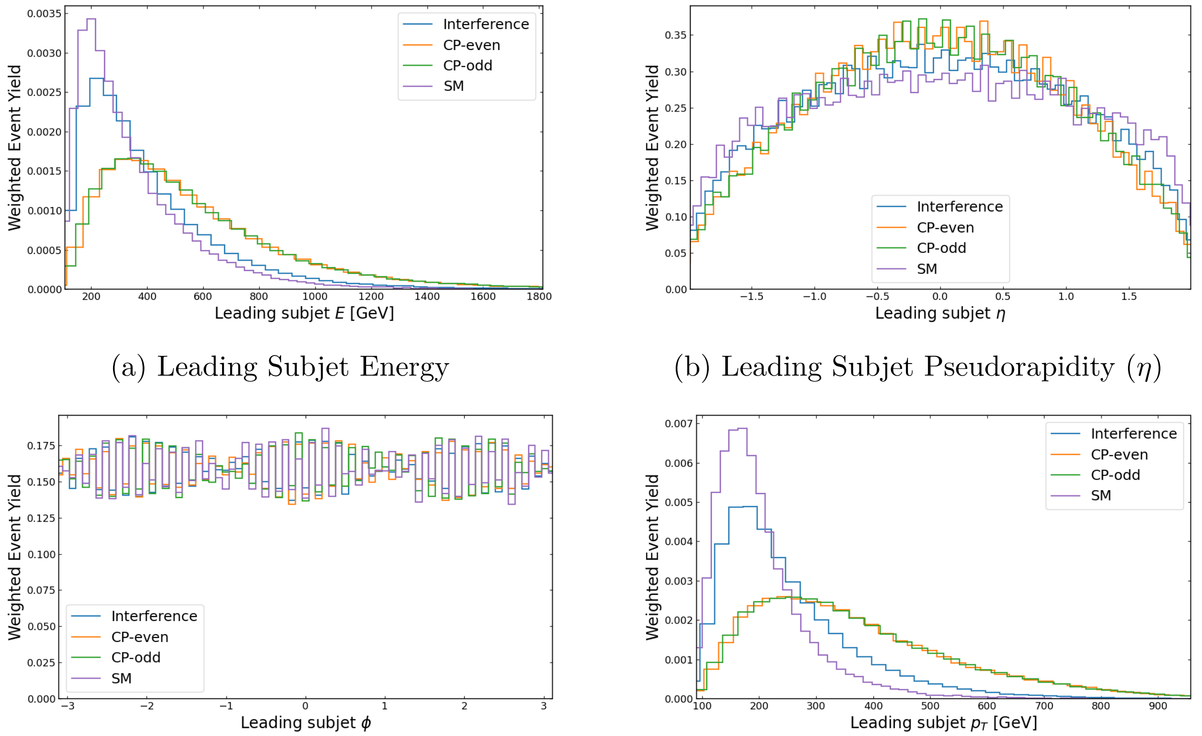
(d) Large-R Jet Transverse Momentum (p_T)



(e) Large-R Jet Azimuthal Angle (ϕ)

Figure 20: Distributions of Large-R jet observables for the four matrix-element components.

Leading Subjet Observables



(a) Leading Subjet Energy

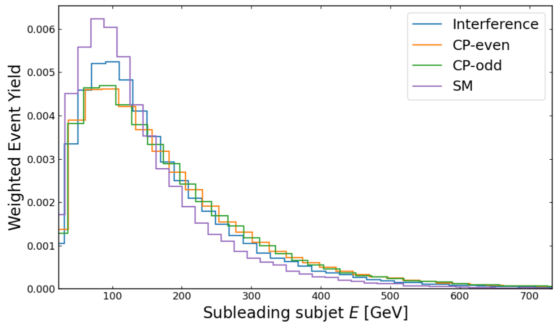
(b) Leading Subjet Pseudorapidity (η)

(c) Leading Subjet Azimuthal Angle (ϕ)

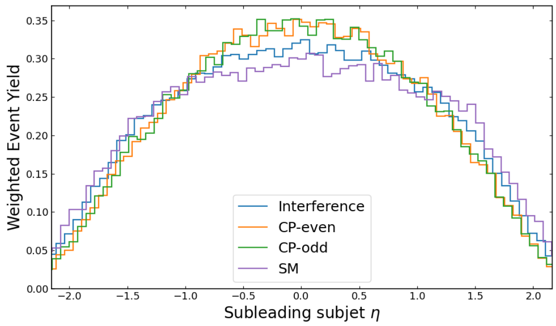
(d) Leading Subjet Transverse Momentum (p_T)

Figure 21: Distributions of leading subjet observables for the four matrix-element components.

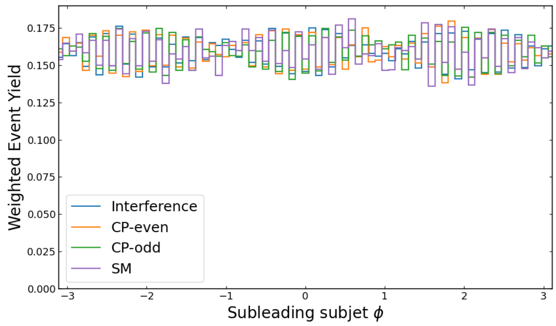
Subleading Subjet Observables



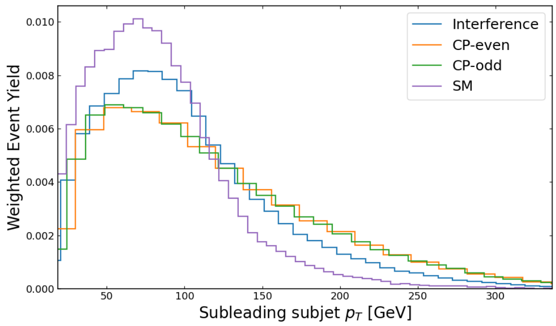
(a) Subleading Subjet Energy



(b) Subleading Subjet Pseudorapidity (η)



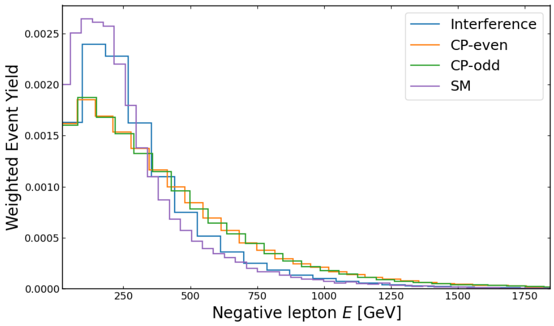
(c) Subleading Subjet Azimuthal Angle (ϕ)



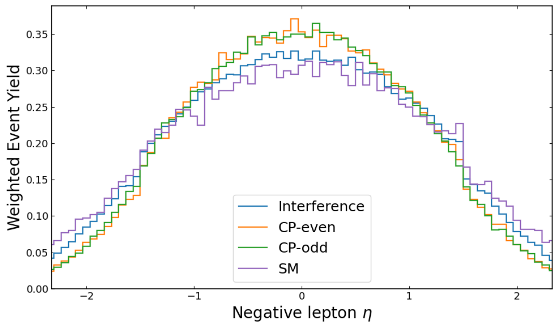
(d) Subleading Subjet Transverse Momentum (p_T)

Figure 22: Distributions of subleading subjet observables for the four matrix-element components.

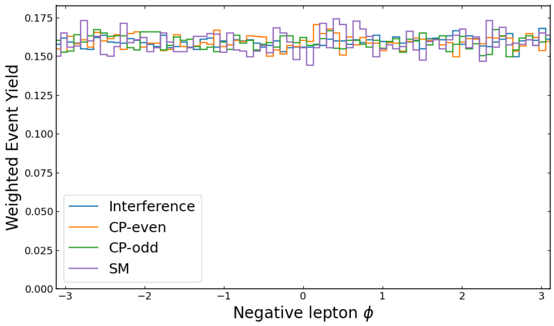
Negative Lepton Observables



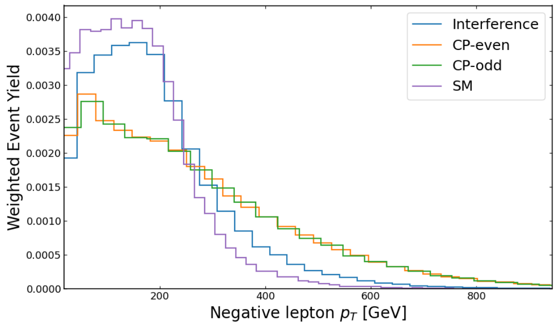
(a) Negative Lepton Energy



(b) Negative Lepton Pseudorapidity (η)



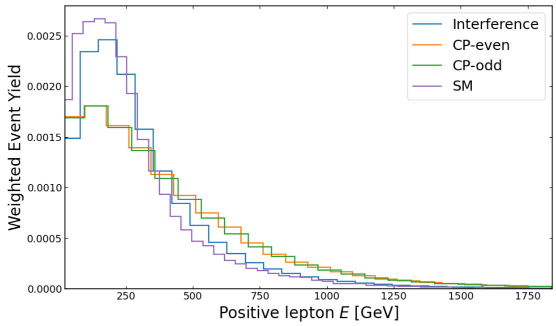
(c) Negative Lepton Azimuthal Angle (ϕ)



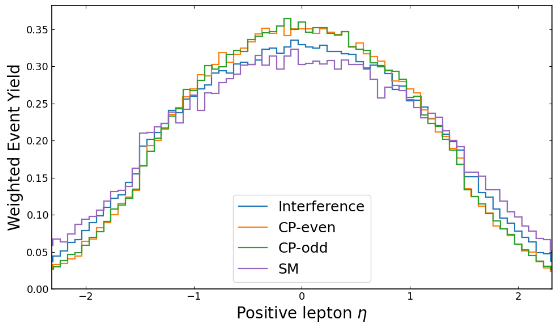
(d) Negative Lepton Transverse Momentum (p_T)

Figure 23: Distributions of negative lepton observables for the four matrix-element components)

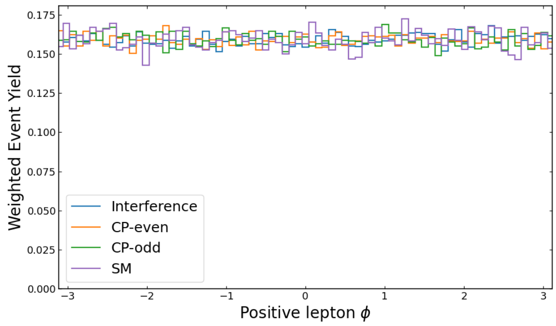
Positive Lepton Observables



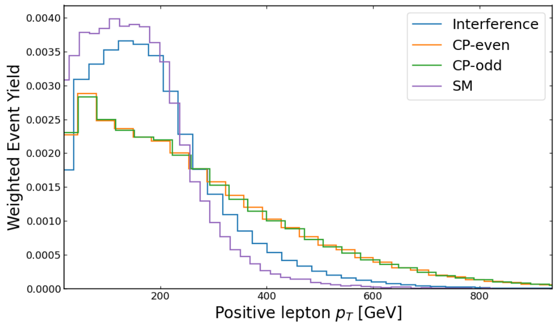
(a) Positive Lepton Energy



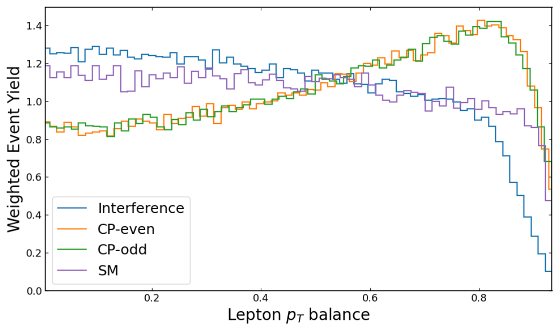
(b) Positive Lepton Pseudorapidity (η)



(c) Positive Lepton Azimuthal Angle (ϕ)



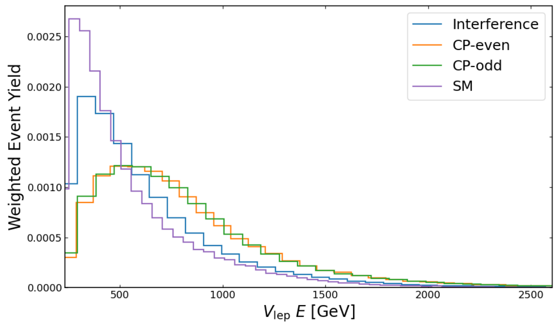
(d) Positive Lepton Transverse Momentum (p_T)



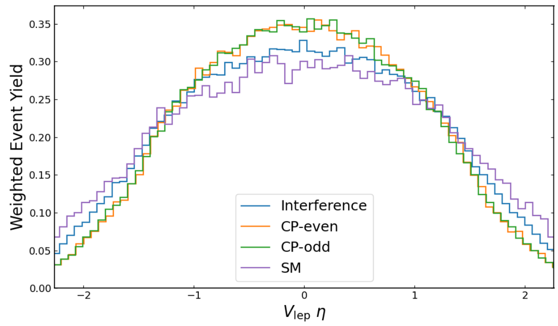
(e) Lepton p_T Balance

Figure 24: Distributions of lepton observables for the four matrix-element components.

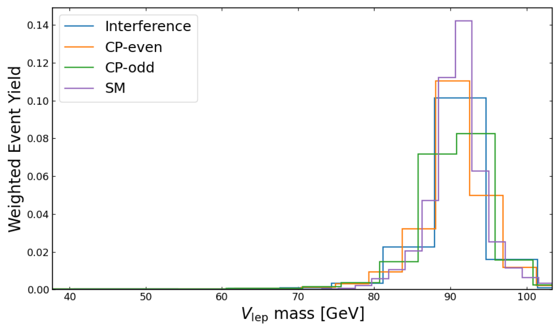
Leptonic Boson Observables



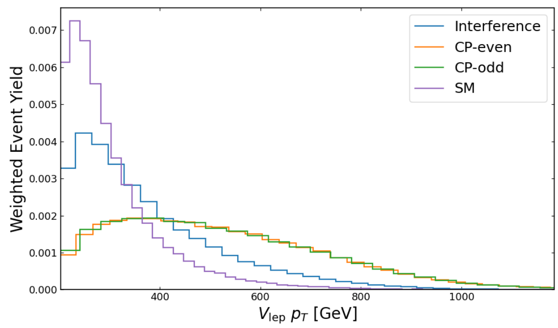
(a) Leptonic Boson Energy



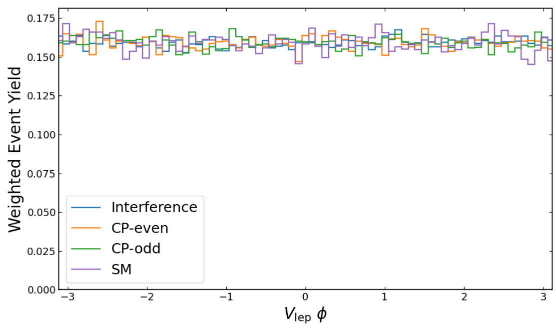
(b) Leptonic Boson Pseudorapidity (η)



(c) Leptonic Boson Mass



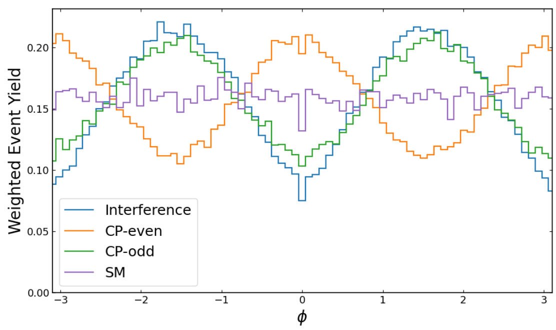
(d) Leptonic Boson Transverse Momentum (p_T)



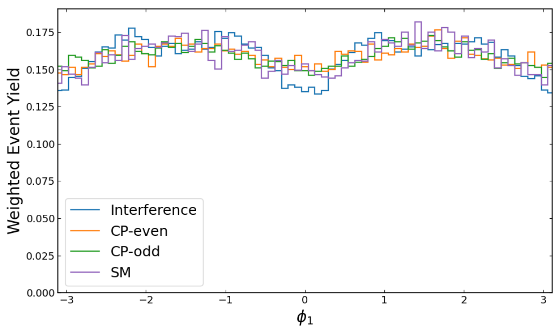
(e) Leptonic Boson Azimuthal Angle (ϕ)

Figure 25: Distributions of leptonic boson observables for the four matrix-element components.

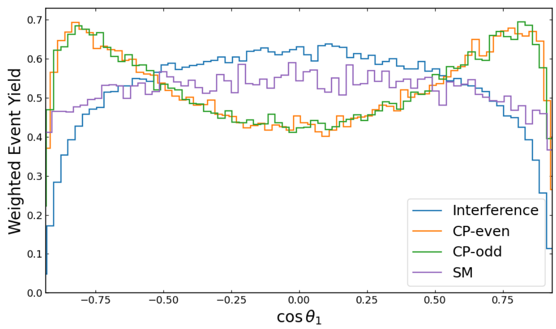
Angular Observables



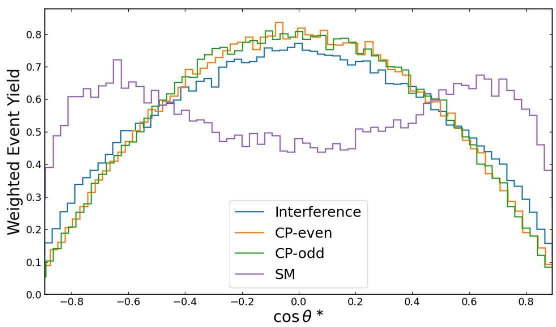
(a) Decay-Plane Angle (ϕ)



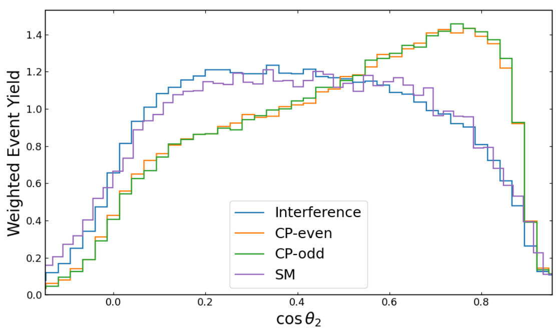
(b) CP-Sensitive Angle (ϕ_1)



(c) $\cos\theta_1$



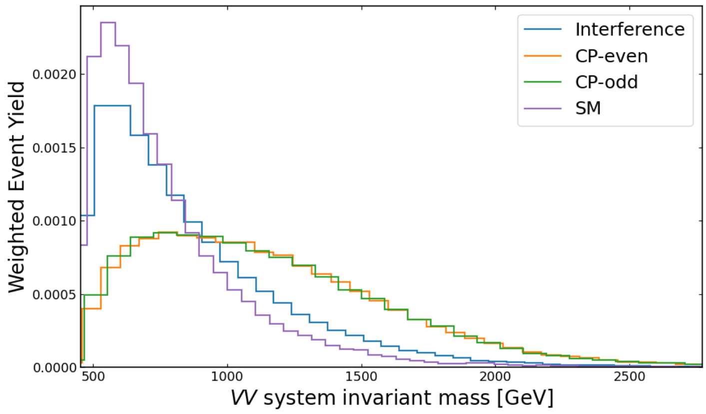
(d) $\cos\theta^*$



(e) $\cos\theta_2$

Figure 26: Distributions of angular observables for the four matrix-element components.

Diboson Invariant Mass



(a) Diboson Invariant Mass (m_{VV})

Figure 27: Distribution of the diboson invariant mass for the four matrix-element components.

A.4 Low-Level Constituent Observables

Table 6: Summary of low-level constituent observables used as input node features for the Graph Neural Network (GNN). These features describe per-particle properties of jet constituents and preserve geometric and kinematic correlations.

Feature	**Description**
constituent_D0	Transverse impact parameter of the particle track
constituent_DZ	Longitudinal impact parameter of the particle track
constituent_E	Energy of the constituent particle
constituent_charge	Electric charge of the constituent particle
constituent_eta	Pseudorapidity of the constituent particle
constituent_isLep	Boolean flag indicating if the constituent is a lepton
constituent_phi	Azimuthal angle of the constituent particle
constituent_pt	Transverse momentum (p_T) of the constituent particle

A.5 High-Level Polynomial Features

Polynomial feature engineering was investigated up to order $N = 2$ for the high-level observables. Before implementation in model training, the mutual information of polynomial features was calculated. The total mutual information increased from 0.0090, with just base observables, to 0.1584, with polynomial features. This demonstrates inclusion of second-order polynomial features results in a mutual information gain of 1754%.

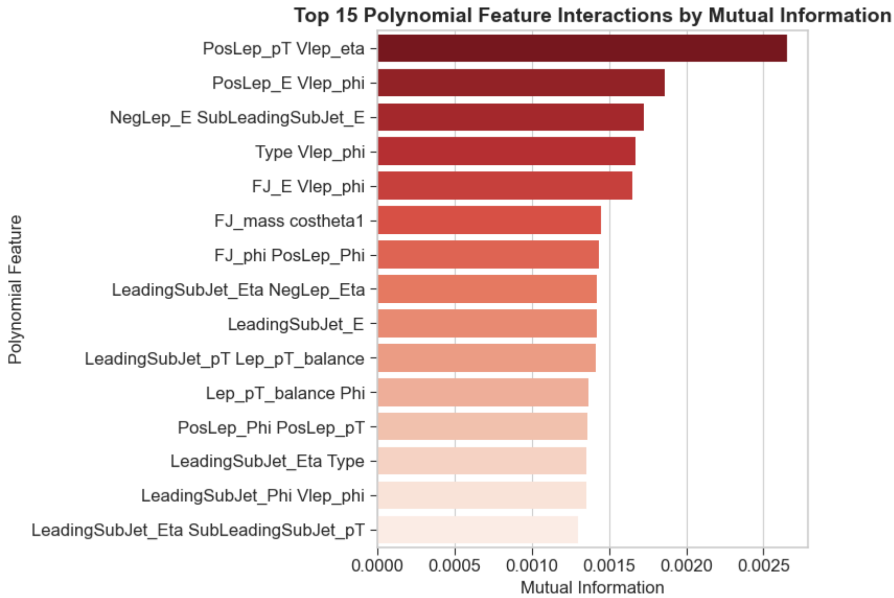


Figure 28: Graph demonstrating the mutual information provided by the top fifteen high-level features, including base and polynomial features. More informative features appear higher on the graph with a deeper shade, less informative features are placed lower with lighter shades.

Figure 28 demonstrates the polynomial and base features ranked by mutual information; the graph demonstrates that the top performing polynomial features carry significantly more mutual information than base features. Figure 29 demonstrates the fraction of total mutual information as a function of the number of features included. It is apparent that base observables rapidly saturate, and a large fraction of mutual information is stored in a few powerful features. On the other hand, polynomial features appear diluted, indicating much of the predictive power in the features is spread across numerous weakly informative features rather than a few strong ones. This hints that polynomial features may dilute the classification task rather than enhance it. Hence, the model was trained on the top 40 features and compared against the model trained on base features.

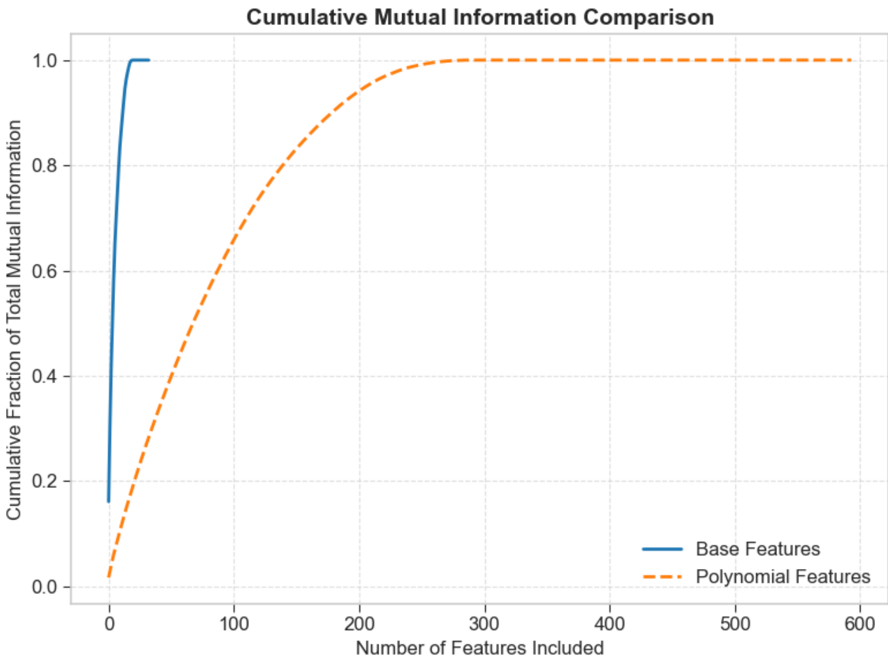


Figure 29: Graph demonstrating the fraction of total mutual information as a function of number of features included. Base level features demonstrate a steeper rise indicating the presence of a small subset of features containing a majority of mutual information while polynomial features demonstrated a more even distribution.

Performance of the DNN with polynomial features resulted in a decrease of AUC score on the order of magnitude $\mathcal{O}(10^{-2})$. Therefore, the final model was trained only including base features.

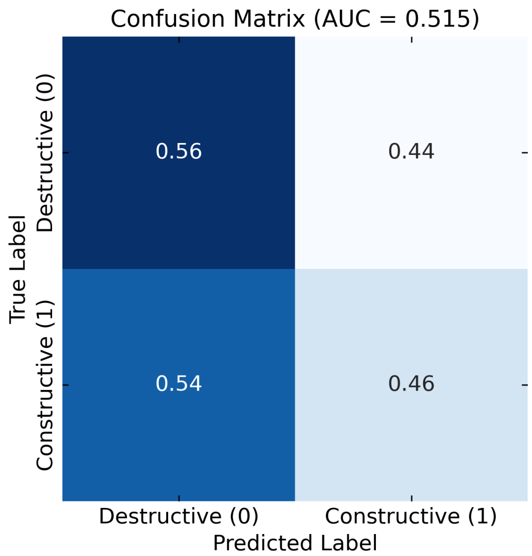
B Model Comparison

Table 7: Summary of model architectures implemented in this work. The DNN processes high-level reconstructed observables, while the GNN operates on low-level constituent features represented as graphs. Both models were trained using the same loss function and optimisation scheme for consistent comparison.

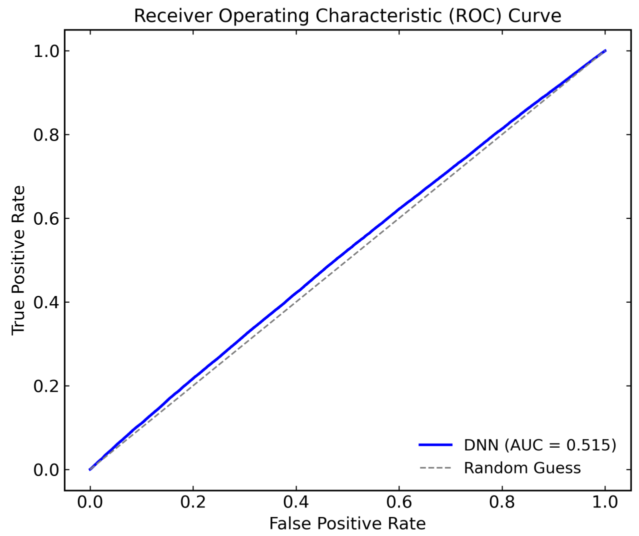
Component	DNN	GNN
Input	18 high-level observables	Up to N_{const} constituent nodes with 8 features each
Feature type	Reconstructed jet and lepton observables	Per-particle kinematics and charges
Input representation	Dense vector	Graph with KNN edges ($k = 6$)
Layer type	Fully connected (Linear)	Graph convolution
Hidden layers	4 ($512 \to 256 \to 128 \to 64$)	3 convolutional ($256 \to 128 \to 64$)
Activations	ReLU	ReLU
Normalization	Batch Normalization	Batch Normalization (per node)
Dropout rate	0.2 (applied after each hidden layer)	0.2 (applied after convolutional layers)
Pooling	–	Global mean pooling (aggregates node embeddings)
Output layer	Linear (1 neuron)	Linear (1 neuron)
Loss function	Binary Cross-Entropy with Logits	Binary Cross-Entropy with Logits
Optimizer	Adam	Adam
Batch size	128	128
Primary metric	ROC–AUC (validation)	ROC–AUC (validation)
Early stopping	Removed after analysis (see Section 3.4.3)	Removed after analysis (see Section 3.4.3)
Implementation	PyTorch 2.8.0	PyTorch Geometric 2.6.1

B.1 DNN Evenly Partitioned vs Oddly Partitioned Results

Figures 30 and 31 contain the confusion matrices and ROC curves for the DNN model trained on even- and odd-indexed event numbers respectively. Figures 30a and 31a indicates the DNN model is able to pick up weakly predictive patterns in the data; however, both models appear to skew towards one class as evidenced by higher performance on predicting destructive interference in Fig. 30a and constructive interference in Fig. 31a. The ROC curves in Figures 30b and 31b further indicate the model learning weakly predictive patterns in the data. Additionally, the similar performance between the even- and odd-indexed models suggests stable generalization and a lack of data leakage.

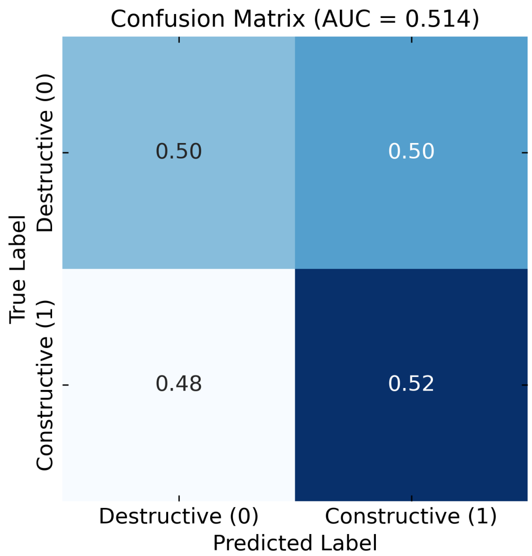


(a) Confusion matrix for model trained on even-indexed event numbers. The model has a good true negative rate of 0.56, and a low false positive rate of 0.44. However, the other metrics are underperforming, highlighting the subtlety in connecting kinematic observables to CP-odd interference.

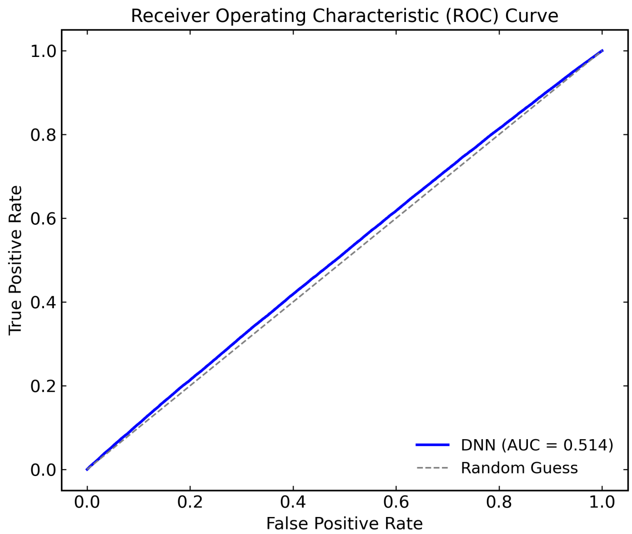


(b) ROC curve for model trained on even-indexed event numbers. The small area above guessing indicates only a weak correlation between the kinematic observables and sign of the interference term.

Figure 30: Confusion Matrix and ROC curve for DNN model trained on even-indexed event numbers. The graphs indicate a slight ability for the model to learn predictive patterns within the high-level observables dataset.



(a) Confusion matrix for model trained on odd-indexed event numbers. The model has a marginally good true positive rate of 0.52, and a low false negative rate of 0.48. However the other metrics are underperforming, highlighting the subtlety in connecting kinematic observables to CP-odd interference.



(b) ROC curve for model trained on odd-indexed event numbers. The small area above guessing indicates only a weak correlation between the kinematic observables and sign of the interference term.

Figure 31: Confusion Matrix and ROC curve for DNN model trained on odd-indexed event numbers. The graphs indicate a slight ability for the model to learn predictive patterns within the high-level observables dataset.

B.2 GNN Results

Three models of GNN were investigated in this work: Graph Convolutional Networks (GCN), a physics-inspired GCN based on the ParticleNet model which excels in Jet Tagging classification tasks [47], and a graph implementation of ResNet which carries a residual of the input layer passed hidden layer outputs [48]. The three models achieved similar performance in classifying the sign of the interference term and thus the GCN was carried forward due to its ease of implementation. The model framework was implemented using `PyTorch_Geometric` [43].

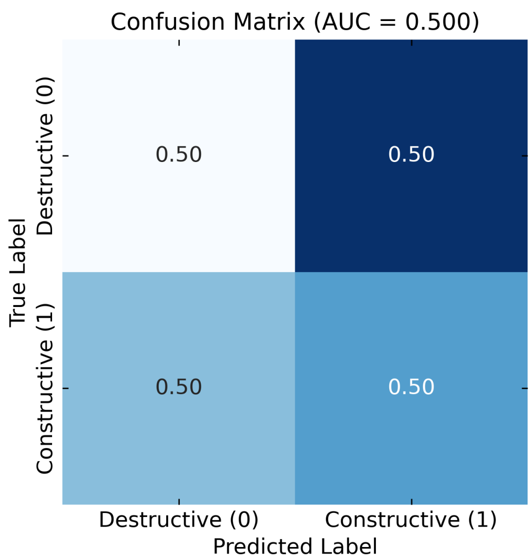
The architecture for the final GCN model consisted of three convolutional layers, each followed by a batch normalization layer, ReLU activation function layer, and a dropout layer during training. Finally, a global pool layer is applied to ensure consistent dimensionality of model outputs. The same early stopping mechanism used for the DNN was employed to mitigate overfitting.

The GNN operated on low-level event information, where each event was represented by a graph where nodes corresponded to individual particle constituents and edges represented pairwise relationships between them. The GNN model was tested on two instances of graph data, one in which all constituents particles were considered and another in which only leptonic particles were considered. The latter dataset was investigated to force the model to classify signal final-states and mitigate constituent noise.

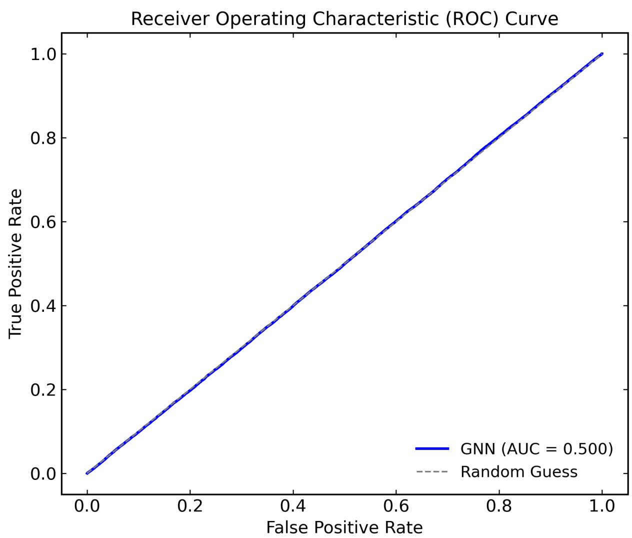
Each node carried the full-set of constituent level features which are detailed in appendix A.4. The lepton-only dataset only contained two nodes, one corresponding to the positively lepton and the other corresponding to the negatively charged lepton, with each node containing 8 feature inputs. The number of constituent particles in each event in the full dataset was inconsistent, with a max number of particles of 60. Therefore, a masking approach was applied to only input features corresponding to a particle. The masking layer checked for positive values of transverse momentum, $constituent\_p_t$, to filter missing input values.

Two different edge index construction methods were applied for the respective datasets. For the lepton only dataset, a fully connected edge index was applied to connect the two leptonic particles. For the full dataset, a $k = 6$ nearest-neighbours approach was used with *constituent_eta* and *constituent_phi*, descriptors of the particles path relative to beam angle, being used to connect particles.

The GCN's performance was compared via AUC score for the lepton-only dataset and the full constituent dataset. Due to a decrease in validation performance, the lepton only dataset was removed from consideration and the full dataset was used going forward. Due to invariance of model performance, the model's hyperparameters were manually tuned. The hidden layer dimension was chosen to be 256 neurons with logarithmic decay, learning rate was set at 0.0001, and the dropout rate was set at 0.2. The GNN was trained for 20 epochs, following the training framework of the DNN.



(a) Confusion matrix for GNN model. The results demonstrate an inability for the GNN to detect any predictive patterns.



(b) ROC curve for GNN model. AUC score of 0.5 indicates an inability for the GNN to detect any predictive patterns.

Figure 32: Confusion Matrix and ROC curve for GNN model. Results indicate the model's absence of predictive power.

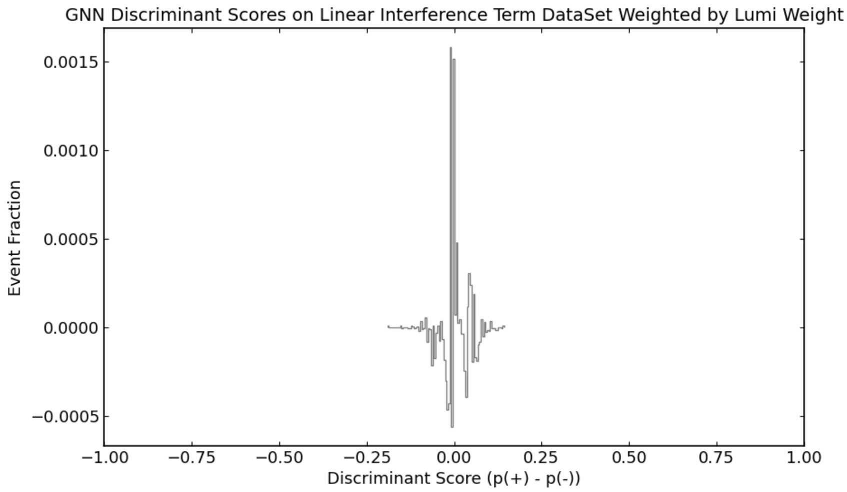


Figure 33: Histogram showing the distribution of interference term sign as a function of the GNN's discriminant score. The counts are weighted by interference term magnitude and sign. The plot shows a lack of (a)symmetry demonstrate the inability of the model to learn any predictive patterns in the data.

Figures 32 and 33 demonstrate the GNN model was unable to learn any meaningful relationships between low-level observables and the sign of the interference term. The performance metrics producing mimic those of random guessing and thus the GNN was not used for final analysis in this work.